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SkinEHDLF a hybrid deep learning approach for accurate skin cancer classification in complex systems

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Skin cancer represents a significant global public health issue, and prompt and precise detection is essential for effective treatment. This study introduces SkinEHDLF, an innovative deep-learning model that enhances skin cancer classification. SkinEHDLF utilizes the advantages of several advanced models, i.e., ConvNeXt, EfficientNetV2, and Swin Transformer, while integrating an adaptive attention-based feature fusion mechanism to enhance the synthesis of acquired features. This hybrid methodology combines ConvNeXt's proficient feature extraction capabilities, EfficientNetV2's scalability, and Swin Transformer's long-range attention mechanisms, resulting in a highly accurate and dependable model. The adaptive attention mechanism dynamically optimizes feature fusion, enabling the model to focus on the most relevant information, enhancing accuracy and reducing false positives. We trained and evaluated SkinEHDLF using the ISIC 2024 dataset, which comprises 401,059 skin lesion images extracted from 3D total-body photography. The dataset is divided into three categories: melanoma, benign lesions, and noncancerous skin anomalies. The findings indicate the superiority of SkinEHDLF compared to current models. In binary skin cancer classification, SkinEHDLF surpassed baseline models, achieving an AUROC of 99.8% and an accuracy of 98.76%. The model attained 98.6% accuracy, 97.9% precision, 97.3% recall, and 99.7% AUROC across all lesion categories in multi-class classification. SkinEHDLF demonstrates a 7.9% enhancement in accuracy and a 28% decrease in false positives, outperforming leading models including ResNet-50, EfficientNet-B3, ViT-B16, and hybrid methodologies such as ResNet-50 + EfficientNet and ViT + CNN, thereby positioning itself as a more precise and reliable solution for automated skin cancer detection. These findings underscore SkinEHDLF's capacity to transform dermatological diagnostics by providing a scalable and accurate method for classifying skin cancer.

Keywords Deep learning, Skin Cancer detection, Hybrid model, ConvNeXt, EfficientNetV2, Swin transformer

Skin cancer ranks among the most prevalent cancers globally, with millions of new cases documented annually. When atypical skin cells proliferate uncontrollably, they produce malignant tumors[1,2]. The three predominant forms of skin cancer are melanoma, basal cell carcinoma (BCC), and squamous cell carcinoma (SCC). Although BCC and SCC are typically less aggressive, melanoma is the most perilous due to its rapid dissemination to other organs if not promptly identified and addressed. The incidence of skin cancer is progressively rising due to heightened exposure to ultraviolet (UV) radiation from both natural and artificial sources, rendering it a significant global health issue[3,4]. Early skin cancer detection increases the chances of survival and effective treatment, improving patient outcomes. Patients who receive timely diagnosis typically have a better prognosis, and many cases can be treated by surgically removing the tumor. Significant management challenges associated with advanced skin cancer result in higher death rates. Conventional detection techniques, such as dermatologists'

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visual assessments and biopsy operations, are time-consuming, arbitrary, and prone to human error. As a result, there is an increasing need for skin cancer detection methods that are more precise, effective, and automated[5–7].

Because they can precisely analyze vast volumes of medical data and images, machine learning (ML) and deep learning (DL) techniques are gaining increasing recognition in skin cancer diagnosis. Modern techniques like CNNs and transfer learning can automate skin cancer detection. Nonetheless, these methodologies encounter further obstacles, including the restricted availability of labeled data, addressing the variability of skin lesion presentations across different skin tones, and minimizing false positives and negatives[8,9]. Confronting these challenges is crucial for developing a reliable and scalable instrument enabling clinicians to make more informed decisions and improve patient care. This study introduces SkinEHDLF, an innovative deep-learning model designed to address these limitations and enhance the accuracy of skin cancer classification[10].

Challenges in skin cancer
The diagnosis of skin cancer presents several difficulties that complicate precise identification[6,11–13].

- Visual Similarity Between Lesions: Several benign and malignant skin cancers exhibit similar visual characteristics, making it challenging for experienced dermatologists to distinguish between them[7].
- Limited Access to Dermatological Expertise: In many rural or resource-constrained regions, access to specialized dermatologists is limited, resulting in delayed diagnoses and poorer patient outcomes.
- High Inter and Intra-Class Variability: Skin lesions can differ significantly in appearance depending on skin types, tones, and imaging conditions, making classification tasks more difficult[8,14].
- Data Imbalance in Existing Datasets: Publicly available skin cancer datasets often exhibit a class imbalance, with more benign lesions than malignant ones, which can skew model training and negatively impact performance[10].
- False Positives and Negatives: Numerous existing deep learning methods and traditional diagnostic methods continue to encounter challenges with misclassification, potentially leading to unnecessary biopsies or, conversely, the overlooking of cancer cases[12,15].

Role of deep learning in skin cancer diagnosis
In recent years, deep learning has emerged as a popular approach to addressing the challenges in skin cancer diagnosis. CNNs have demonstrated exceptional efficacy in medical image analysis, including skin cancer detection. These networks have proven highly effective, sometimes outperforming human experts in accuracy[5–7]. However, CNN-based architectures have issues with feature extraction and generalization across lesions. Vision Transformers (ViTs) have gained popularity as a solution to these limitations due to their ability to utilize self-attention mechanisms to identify both local and global dependencies in images. Skin cancer detection is one of the numerous image recognition tasks where these models have demonstrated promise in improving accuracy[4,14,16,17]. CNNs and transformers are combined in hybrid models, which have emerged as recent trends in the field. Combining the best aspects of both approaches, these hybrid models aim to provide a more reliable classification system for skin cancer[2].

Research target and motivations
The primary cause of cancer-related deaths globally is still skin cancer, and improving survival rates requires early detection. Skin cancer detection remains challenging despite significant advancements in medical imaging and diagnostic tools, primarily due to the wide range of lesion appearances, the subtle differences between benign and malignant growths, and the difficulty in accurately interpreting these variations[18]. Existing automated systems frequently fail to generalize across different datasets, skin types, and image qualities, resulting in misclassifications that can lead to unnecessary treatments or missed diagnoses. The motivation for this study stems from the need to enhance the accuracy and reliability of automated skin cancer detection, particularly given the limited access to specialized dermatological expertise and the increasing demand for scalable solutions. Traditional deep learning models, while promising, often rely on a single architecture, which may fail to capture the full range of features necessary for precise classification. This study combines the strengths of several cutting-edge deep learning models, EfficientNetV2, ConvNeXt, and Swin Transformer, to create a more robust, adaptable, and high-performing framework. Each of these models exhibits various strengths[8,19].

EfficientNetV2 is renowned for its efficiency and high performance with limited computational resources. ConvNeXt excels at feature extraction, capturing fine details from complex images, while the Swin Transformer utilizes self-attention mechanisms to capture both local and global patterns. Integrating these architectures into a cohesive system aims to leverage their complementary advantages, enabling the model to address the complexities of skin cancer classification more effectively[2,14]. An adaptive feature fusion mechanism is proposed to enhance classification accuracy. This approach enables the model to dynamically optimize feature selection, guaranteeing that only the most pertinent information is considered during decision-making. This research aims to develop a system capable of accurately, reliably, and scalable detecting skin cancer while tackling the core issues of misclassification, dataset imbalance, and inadequate generalization. The system can practically be applied in clinical and resource-limited environments[4,7,8,10,15].

Key research questions
This research seeks to answer the following key questions[9,18–21].

- How can a hybrid deep learning framework improve the accuracy and robustness of binary skin cancer classification?

- What advantages can be gained from combining CNN-based architectures with transformer models in medical image analysis?
- How might adaptive feature fusion methods improve lesion representation and classification accuracy?
- Is it possible for the proposed E-HDLF model to surpass the performance of current deep learning methods, including ResNet-50, EfficientNet-B3, and ViT-B16, in terms of accuracy, precision, and AUROC?

Key contributions

The E-HDLF framework is designed to enhance skin cancer detection accuracy by overcoming current methods' constraints, thereby offering a dependable, scalable, and durable solution that can be implemented in real-world clinical environments. We anticipate that these developments will enhance the ongoing effort to improve the early detection of skin cancer, leading to more effective healthcare delivery and improved patient outcomes[2,16,17]. The following contributions are presented in the present research.

- *Development of E-HDLF*: A novel hybrid deep learning framework that integrates CNNs and transformers for more accurate and reliable lesion classification.
- *Adaptive Attention Fusion*: An adaptive feature fusion mechanism that dynamically prioritizes features based on lesion complexity has been introduced, improving model performance.
- *Transfer Learning and Fine-Tuning*: To fine-tune the model, transfer learning strategies are applied to the ISIC 2024 dataset, comprising various skin lesion images.
- *Comprehensive Evaluation*: A thorough comparative analysis and comparison with existing models, including ResNet-50, EfficientNet-B3, ViT-B16, ResNet-50 + EfficientNet, and ViT + CNN, demonstrate that the proposed model significantly outperforms baseline architectures in key performance metrics.
- *Real-World Impact*: This paper explores the potential implications of incorporating artificial intelligence-driven skin cancer diagnostics into clinical settings and highlights the need for further validation for practical applications.

Organization of the paper

The remainder of the paper is structured as follows:

- *Section 2: Literature Review*. This section provides a comprehensive review of current deep-learning methods for skin cancer classification, detailing their strengths, weaknesses, and areas for improvement.
- *Section 3: Materials & Methods* This section elucidates the architecture of the E-HDLF framework, outlining the dataset preprocessing procedures, model design, and training methodologies employed to enhance performance.
- *Section 4: Experimental Results and Discussion*. This section presents our model's performance outcomes, along with assessment metrics such as AUROC, recall, accuracy, and precision. In addition to ablation studies, a comparison with current models will be presented.
- *Section 5: Conclusion and Future Directions*. In the final section, we will highlight the main conclusions, discuss potential practical applications of our model in clinical settings, and suggest future research directions for enhancing deep learning-based skin cancer detection.

Related work

Skin cancer classification has advanced significantly since introducing hybrid deep learning models that combine CNNs, transformers, RNNs, attention mechanisms, and optimization strategies. This section categorizes the current literature into three groups based on the hybrid techniques employed to enhance diagnostic accuracy, generalizability, and real-time application.

Based on hybrid CNN-transformer models for skin cancer classification

Several studies have investigated hybrid models that combine CNNs and Transformers to improve skin cancer classification, utilizing the benefits of both architectures. Pacal et al.[22] presented a hybrid CNN-Vision Transformer (ViT) model. ViTs capture global contextual relationships, while CNNs extract low-level spatial features. The study aimed to enhance multi-class skin cancer classification, specifically the differentiation between melanoma, basal cell carcinoma, and squamous cell carcinoma. They improved feature extraction with spatial attention mechanisms, and the model achieved 94.5% accuracy on the ISIC-2020 and HAM10000 datasets, outperforming standalone CNNs. ViTs, on the other hand, require a significant number of computational resources and training data, resulting in a substantial processing overhead.

Natha et al.[6] investigated skin cancer detection with an ensemble learning strategy that included CNNs and Transformers. Their hybrid technique used multiple-scale feature extraction to generate fine-grained and high-level lesion data. The methodology increased the F1-score by 6% compared to traditional CNN models; however, the model's interpretability is poor, making it difficult for clinicians to understand the decision-making process. Kandhro et al.[17] introduced the Enhanced VGG19 (E-VGG19) model, which combines transformer layers with VGG-based CNN architectures. This model achieved an accuracy of 96.2% when applied to dermoscopic images, and it was designed to detect skin cancer in real-time with no delay. It performs noticeably worse than external data because it struggles to apply to previously unseen datasets.

Ahmed et al.[18] examined a hybrid CNN-transformer methodology for melanoma lesion segmentation and classification, integrating multi-scale CNN feature extraction with transformer-based self-attention mechanisms. They reported a 2.5% increase in sensitivity compared to previous models but had difficulty detecting small lesions because transformer models require more significant areas of lesions for effective segmentation. Almuayqil et al.[19] proposed a multi-type feature fusion model that combines CNN and transformer-based feature extraction to classify multiple skin lesions. This method enhanced the precision of rare skin cancer types; however, the longer inference times present challenges for real-time deployment. Although hybrid CNN-transformer models have exhibited exceptional classification performance, their high computational cost remains a substantial impediment. Future research will likely concentrate on developing lighter transformer models that can be implemented in clinical environments with restricted resources.

Based on hybrid CNN-RNN and attention-based models for skin lesion analysis

Other researchers have taken a different approach, combining CNNs with RNNs or attention mechanisms to enhance lesion detection. Mavaddati[3] developed a CNN-LSTM hybrid model that utilizes CNNs for feature extraction and LSTM networks for temporal sequence learning. This model is particularly effective for tracking temporal alterations in melanoma lesions. It achieved an accuracy of 92.8% using sequential images. The primary drawback is that LSTMs exhibit computational inefficiency, resulting in prolonged training times.

Zareen et al.[10] introduced a hybrid model that combines CNNs for feature extraction and RNNs for capturing sequential dependencies. Their model outperformed standalone CNNs by 4.5% in classification accuracy. Nonetheless, RNNs require extensive datasets for optimal performance, which limits their applicability in smaller datasets. Omeroglu et al.[23] introduced a soft attention-driven deep learning framework for the multi-label classification of skin lesions. This model utilizes attention mechanisms to emphasize the essential characteristics of lesions while ignoring extraneous information. The research indicated a 5% enhancement in specificity, rendering it more suitable for clinical application. Nonetheless, attention mechanisms require extensive hyperparameter optimization, complicating model deployment.

Akram et al.[24] introduced a dual-attention CNN model that utilizes soft and hard attention mechanisms to enhance lesion localization and feature selection. This model enhanced segmentation accuracy to 89.3% but consumed more memory, complicating deployment on mobile devices or edge computing systems. Although hybrid CNN-RNN and attention-based models enhance sequential learning and lesion localization, their computational complexity and dependence on massive datasets make real-world implementation challenging.

Based on hybrid feature engineering and optimization-based models

A separate research category examines hybrid models integrating deep learning with feature engineering and optimization methods to enhance skin cancer classification. Farea et al.[8] presented a hybrid deep-learning model incorporating convolutional neural networks with manually crafted texture features to classify melanoma. This method resulted in a notable enhancement in sensitivity rates (94.1%). The dependence on manual feature selection limits scalability and requires domain expertise for practical feature engineering.

Packiamary and Muthukumaravel[16] developed a multimodal deep learning model that integrates feature engineering methods with CNN architectures. This model attained superior accuracy on multimodal datasets, indicating potential for clinical applications. The primary constraint, however, was the necessity for extensively annotated datasets, which are frequently limited in medical imaging. Kumar Lilhore et al.[2] presented a hybrid model that combines U-Net and MobileNet-V3, employing hyperparameter optimization techniques to improve classification accuracy. The methodology demonstrated superiority over traditional U-Net models, attaining a segmentation accuracy of 91.5%. This model requires significant computational resources, which presents challenges for real-time implementation. However, models that combine.

feature engineering and other techniques require considerable manual adjustment. These models improve both generalization and interpretability. Subsequent research should concentrate on automated feature selection methodologies to enhance efficiency and augment scalability. The reviewed studies indicate that hybrid models, whether utilizing CNN-Transformer, CNN-RNN, or feature engineering techniques, significantly enhance skin cancer classification. Nonetheless, obstacles persist, particularly regarding computational complexity, model interpretability, and the need for extensively annotated datasets. Future research should focus on developing more efficient and generalizable models suitable for real-time clinical applications with minimal computational requirements. Table 1 presents a comparative analysis of various existing research based on numerous factors.

Materials and methods

Dataset details

The ISIC 2024 dataset is recognized as one of the most accessible collections for skin lesion analysis, containing over 401,059 high-resolution images[25,26]. These images are derived from 3D total body photography, providing a more precise and clinically relevant representation of skin lesions compared to traditional dermoscopic images. This feature enhances the dataset's value for deep learning model development by incorporating authentic variations in lesion morphology, patient demographics, and environmental factors.

The dataset encompasses a diverse range of Fitzpatrick skin types (Types I to VI), making it applicable to various ethnic backgrounds. The classification covers multiple types of skin lesions, including melanoma, basal cell carcinoma, squamous cell carcinoma, and benign nevi. Every category is adequately represented, with a roughly equal distribution of malignant and benign lesions to minimize bias during model training. The dataset included images that were validated through histopathological or clinical methods. We excluded lesions that were captured in low-light or had too many obstructions to maintain high image quality. Furthermore, we eliminated duplicate images and samples that were not suitable for diagnosis during the preprocessing stage.

The dataset encompasses a range of skin lesion types, various lighting conditions, and patient characteristics, including age, gender, and ethnicity, thereby providing a comprehensive representation of scenarios commonly encountered in clinical settings[25,27–29]. To improve the model's reliability, techniques such as histogram equalization, contrast normalization, and augmentation methods (including rotation, flipping, and scaling) were employed. The collection includes images of benign and malignant skin cancer, each sized at 224 by 224

Study	Hybrid CNN & Transformer	Hybrid CNN & LSTM	Multi-modal Imaging	Feature Fusion	Adaptive Attention	High Accuracy	Computational Cost	Generalizability	Model for Real-time	Multi-class Classification	Dataset Used
Ozdemir & Pacal[1]	Yes	No	No	Yes	No	Yes	High	High	No	No	Yes
Mavaddati[2]	No	Yes	No	No	No	Yes	Moderate	High	No	No	Yes
Pacal et al.[3]	Yes	No	No	No	Yes	Yes	High	High	No	No	Yes
Natha et al.[4]	Yes	No	No	No	Yes	Yes	Moderate	High	Yes	No	Yes
Jeyageetha et al.[5]	Yes	No	Yes	Yes	No	Yes	High	Moderate	Yes	No	No
Farea et al.[6]	Yes	No	No	Yes	No	Yes	High	High	No	No	No
Zareen et al.[7]	Yes	Yes	No	No	Yes	Yes	High	High	Yes	No	No
Ghosh et al.[8]	Yes	No	No	No	No	Yes	Moderate	Moderate	No	No	Yes
Subhashini & Chandrasekar[9]	Yes	No	No	Yes	Yes	Yes	High	High	No	No	No
TY et al.[10]	Yes	No	No	No	Yes	Yes	High	Moderate	No	Yes	Yes
Packiamary & Muthukumaravel[11]	Yes	No	Yes	Yes	Yes	Yes	High	High	Yes	No	No
Kandhro et al.[12]	Yes	No	No	No	Yes	Yes	High	Moderate	Yes	No	No
Kumar Lilhore et al.[13]	Yes	No	No	Yes	Yes	Yes	High	High	No	No	Yes
Ahmed et al.[14]	Yes	No	No	Yes	Yes	Yes	Moderate	High	No	No	No
Bassel et al.[15]	Yes	No	No	Yes	Yes	Yes	High	High	Yes	Yes	No
Mateen et al.[16]	Yes	No	No	Yes	Yes	Yes	High	Moderate	No	No	Yes
Maniraj & Maran[17]	Yes	No	No	No	Yes	Yes	High	Moderate	Yes	No	Yes
Almuayqil et al.[18]	Yes	No	Yes	Yes	Yes	Yes	Moderate	High	Yes	Yes	No
Akram et al.[19]	Yes	No	No	Yes	Yes	Yes	High	Moderate	Yes	Yes	No
Proposed Model (SkinEHDLF)	Yes	Yes	Yes	Yes	Yes	Yes	High	Low	Yes	Yes	Yes

Table 1. Comparative analysis of various existing research in the field of skin cancer research.

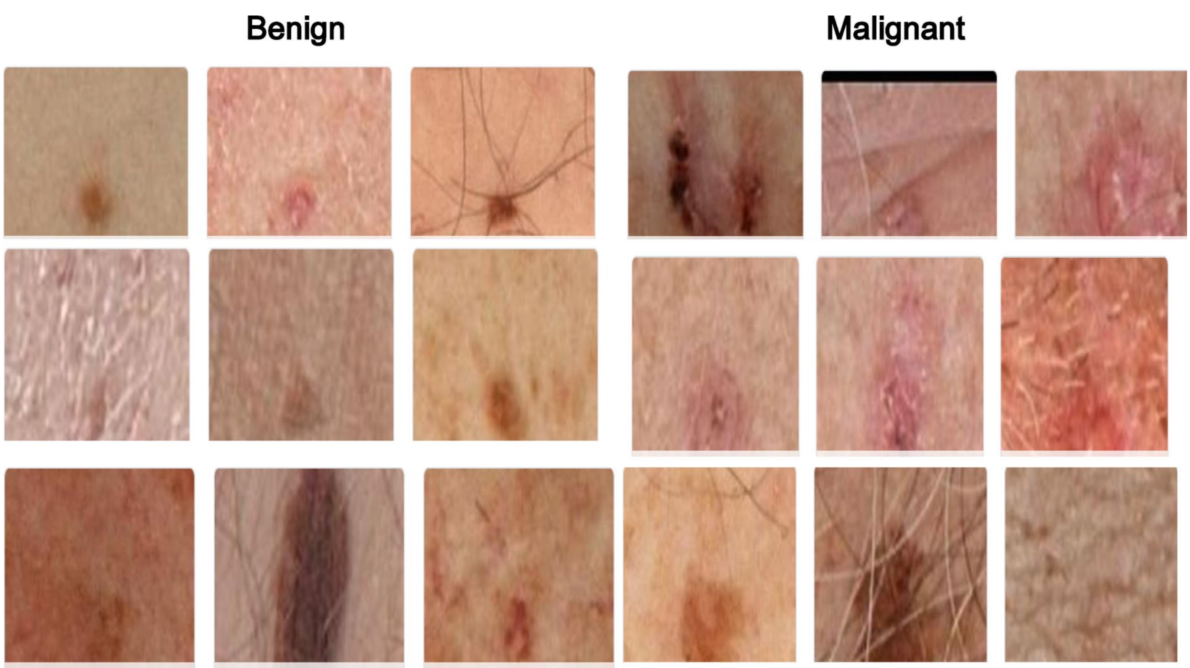

Fig. 1. Overview of image category in the ISIC-2024 dataset.

pixels. These additions ensure that the dataset encompasses a broad spectrum of real-world variations, thereby enhancing its applicability for clinical use. Figure 1 presents the image category in the ISIC-2024 dataset.

The ISIC 2024 dataset supports binary and multi-class classification tasks. The binary classification task involves identifying melanoma by distinguishing between benign (non-cancerous) and malignant (cancerous) lesions. A dataset of skin lesions classified as melanoma, BCC, and SCC is employed in the multi-class classification task, which enables a more granular classification. The dataset is an exceptional resource for developing and testing

Category	Overall Count	Athens	Barcelona	Basel	Brisbane	New York	Sydney	Vienna
Total Images	401,059	7,976	105,724	65,218	51,768	129,068	28,665	12,640
Unique Lesions	401,059	7,976	105,724	65,218	51,768	129,068	28,665	12,640
Manual Tags	22,058	167	1,720	3,004	9,056	6,212	1,792	107
Diagnosis - Malignant	393	6	72	13	81	174	33	14
Diagnosis - Indeterminate	114	1	12	2	60	39	—	—
Diagnosis - Benign	400,552	7,969	105,640	65,203	51,627	128,855	28,632	12,626
Anatomic Site - Head/Neck	12,046	353	3,416	2,320	1,533	3,229	809	386
Anatomic Site - Anterior Torso	87,770	1,774	23,546	14,075	7,806	31,525	5,722	3,322
Anatomic Site - Posterior Torso	121,902	2,842	32,725	19,822	13,569	40,495	8,366	4,083
Anatomic Site - Upper Extremity	70,557	1,475	18,536	11,312	9,676	22,225	5,310	2,023
Anatomic Site - Lower Extremity	103,028	1,532	27,332	15,600	16,841	30,441	8,458	2,824
Anatomic Site - Unknown	5,756	—	169	2,089	2,343	1,153	—	2
Lighting - White Light	115,156	—	1	484	126	114,545	—	—
Lighting - XP Light	285,903	7,976	105,723	64,734	51,642	14,523	28,665	12,640

Table 2. ISIC 2024 dataset overview based on cancer classes and tags[25,29].

Category	Overall Count	Athens	Barcelona	Basel	Brisbane	New York	Sydney	Vienna
Total Patients	1,042	16	163	230	176	398	44	15
Male Patients	551	9	75	115	87	229	26	10
Female Patients	458	7	87	88	84	169	18	5
Unknown Sex	33	—	1	27	5	—	—	—
Age Group [− 20)	8	—	—	6	1	1	—	—
Age Group [20–25)	16	1	2	5	1	7	—	—
Age Group [25–30)	28	—	4	5	2	17	—	—
Age Group [30–35)	56	2	4	16	10	23	1	—
Age Group [35–40)	59	—	3	21	11	22	1	1
Age Group [40–45)	107	2	10	24	20	43	7	1
Age Group [45–50)	85	—	17	20	16	27	4	1
Age Group [50–55)	119	5	26	23	17	40	5	3
Age Group [55–60)	133	2	16	35	26	48	4	2
Age Group [60–65)	122	2	21	21	21	46	9	2
Age Group [65–70)	129	—	18	16	32	56	5	2
Age Group [70–75)	76	1	12	18	12	29	3	1
Age Group [75–80)	46	—	12	9	1	21	3	—
Age Group [80–85)	33	1	10	5	—	14	2	1
Age Group [85+)	12	—	4	2	1	4	—	1
Unknown Age	13	—	4	4	5	—	—	—

Table 3. Presents an overview of the dataset based on patient ages.

deep learning models for automated skin cancer detection in clinical settings due to its high-quality images, large size, and diversity. Table 2 provides a detailed overview of the ISIC 2024 dataset, organized by cancer classes and tags, while Table 3 presents an overview of the dataset based on patient age groups.

Architecture of proposed SkinEHDLF hybrid model

The SkinEHDLF model combines three robust deep-learning architectures: ConvNeXt, EfficientNetV2, and SwinTransformer. Each model offers a distinct set of benefits in terms of feature extraction, image classification, and understanding of contextual information. These benefits are then incorporated into the system through an adaptive attention-based feature fusion mechanism[19,21,23,24]. This hybrid method encompasses various skin lesion characteristics while enhancing classification accuracy for skin cancer detection.

Table 4 presents the Proposed SkinEHDLF Architecture Configuration. The SkinEHDLF model's structure explains how each component is critical to detecting skin cancer. The process begins with the Input Layer, which takes images of skin lesions as input for the model's analysis. Next, the model utilizes three robust networks: ConvNeXt, EfficientNetV2, and Swin Transformer, each offering distinct advantages, such as fast and efficient feature extraction, adaptability to varying data sizes, and the ability to capture long-range dependencies in the image[11–13]. The Feature Fusion Layer subsequently integrates the essential features from all three networks,

Layer	Type	No. of Units	Activation Function	Kernel Size	Stride
Input Layer	Image Input	–	–	–	–
Feature Extraction (ConvNeXt)	Convolutional + Attention	Varies	ReLU	3 × 3	1
Feature Extraction (EfficientNetV2)	Convolutional + Attention	Varies	Swish	3 × 3	2
Feature Extraction (Swin Transformer)	Self-Attention + Convolutional	Varies	GELU	4 × 4	2
Feature Fusion Layer	Fusion of Extracted Features	–	–	–	–
Fully Connected (Dense) Layer	Dense Layer	1024	ReLU	–	–
Final Output Layer	Softmax / Sigmoid (depending on task)	2 (binary) or N (multi-class)	Softmax (multi-class) / Sigmoid (binary)	1 × 1	1

Table 4. Proposed SkinEHDLF architecture configuration.

guaranteeing the amalgamation of the optimal characteristics of each model. The Fully Connected (dense) layer analyzes the integrated features, allowing the model to identify intricate patterns and nuanced distinctions among images. The Output Layer ultimately renders the final prediction, categorizing the lesions as benign or malignant, using either SoftMax or sigmoid activation functions, depending on the specific task. This design ensures that the model can accurately evaluate skin lesions, making it an effective identification tool for skin cancer. Figure 2 presents the architecture of the proposed SkinEHDLF hybrid model.

Working of proposed SkinEHDLF hybrid model
The proposed SkinEHDLF Hybrid Model includes the following key concepts.

Input layer
The SkinEHDLF Hybrid Model's Input Layer accepts and prepares raw skin lesion images for processing. This layer is responsible for image acquisition. Advanced imaging technologies, including 3D total body photography (TBP), capture high-resolution images of skin lesions[25,26,29].

Data pre-processing layer
The SkinEHDLF Hybrid Model's Data Pre-Processing Layer is crucial for improving the relevance of input data before incorporating it into the deep learning architecture. This layer converts unprocessed skin lesion images into a format suitable for feature extraction and classification.

<u>Data augmentation</u> To enhance the diversity of the training data and mitigate the risk of overfitting, several data augmentation techniques were employed. These augmentations replicate authentic variations in image quality and capture conditions, guaranteeing that the model acquires resilient features that generalize effectively. The subsequent augmentation strategies were utilized:

- Rotation: Images were randomly rotated within a specified range of ± 30° to replicate diverse orientations of the skin lesions.
- Scaling and Cropping: The images underwent random scaling, followed by random cropping to a fixed size of 224 × 224 pixels. This step ensured that the input dimensions were consistent while also accounting for minor variations in the appearance of the lesions.
- Flipping: Horizontal and vertical flips were applied to the images, increasing the model's ability to recognize lesions regardless of their orientation.
- Color Jittering: Random modifications to the brightness, contrast, and saturation of the images were executed, emulating variations in lighting and camera settings.
- Gaussian Noise: Small amounts of random noise were added to the images to improve the model's robustness to noisy inputs, which are common in real-world image data.
- These augmentation techniques helped to artificially expand the dataset and provide more variety, allowing the model to learn better and avoid overfitting to specific image patterns.

<u>Normalization and standardization</u> To optimize the learning process and improve the model's ability to train efficiently, all images were normalized. Pixel values were scaled to a range of 0 to 1 by dividing them by 255 (standard normalization). Additionally, mean subtraction and standard deviation normalization were performed using the dataset's global mean and standard deviation. This step ensured that the pixel values had zero mean and unit variance, allowing deep learning models to converge more quickly during training.

<u>Handling class imbalance</u> Given the imbalanced nature of the skin lesion dataset, where specific categories (e.g., melanoma) are underrepresented, it was critical to address this issue to avoid biased model predictions. To address this, the following strategies were used:

- Class-weighted loss function: We used a class-weighted loss function during training, which assigns higher weights to the underrepresented classes (melanoma) and lower weights to overrepresented classes. This approach enabled the model to pay more attention to underrepresented categories without requiring changes to the dataset itself.

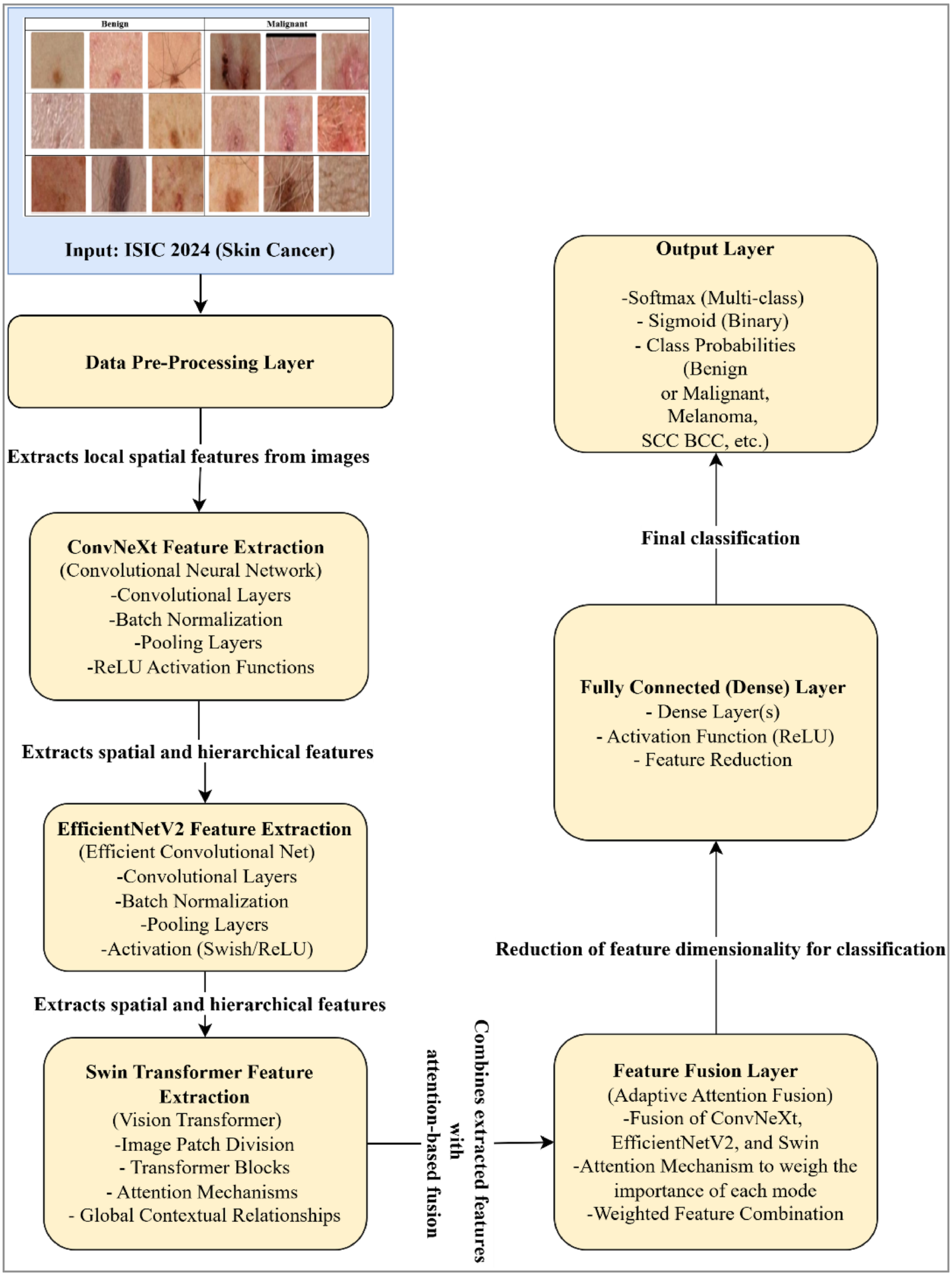

Fig. 2. Architecture of the proposed SkinEHDLF hybrid model.

- Data augmentation tailored to classes: Augmentation techniques were applied to the minority classes, especially melanoma, to artificially increase the number of samples and balance the class distribution. This was accomplished while ensuring that overall data integrity was maintained.

These methods reduced the potential bias caused by class imbalances, ensuring that the model's performance was not skewed in favor of the majority class.

Image resizing and resolution adjustment The images in the ISIC 2024 dataset came in varying resolutions. To standardize the input size and meet the input requirements of the deep learning models (ConvNeXt, EfficientNetV2, and Swin Transformer), all images were resized to 224 × 224 pixels. This resizing process reduced computational costs and ensured that the model could process all photos consistently, which is crucial for successful training.

Efficient preprocessing of skin lesion images is crucial for the SkinEHDLF model to extract significant features and generate accurate predictions. This study employed various preprocessing techniques to augment dataset quality, enhance model performance, and mitigate overfitting. The following are the essential steps executed in the data preprocessing pipeline. Algorithm 1 presents the algorithm for data preprocessing in the proposed SkinEHDLF Hybrid Model. Table 5 presents an overview of the dataset after the data pre-processing phase.

Input: Input raw skin cancer images (ISIC-2024) I_{Raw}
Output: Pre-processed iages I_{Final}.
Step 1: Resize Images. 1.1 To maintain the uniformity among cancer images, change the size of all the skin cancer images into $(H \times W)$ With a size of (224×224) Pixels, using Eq. (1). Here I_{Resize}: Image resized, F_{Resize}: Resizing function, H: Height and W: Width. $I_{Resized} = F_{Resize}\,(I_{Raw},\ H,\ W)$ (1)
Step 2: Normalization process. *2.1 To normalize the pixel values to a range of [0, 1], divide by the maximum pixel value (255 for 8-bit images), Eq. 2. Here* $I_{Normalized}$ *Normalized images have pixel values ranging from 0 to 255, converted to values from 0 to 1.* $I_{Normalized} = \frac{I_{Resized}}{255}$ (2)
Step 3: Apply Data Augmentations. *3.1 To make the dataset more diverse, use augmentation techniques such as flipping, rotating, zooming, or shifting, as described in Eq. 3. Here* $I_{Augmented}$: *Augmented image,* $F_{Augmented}$: *Augmented function, which includes Rotation, Flipping, Zoom, and Translation operations)* $I_{Augmented} = F_{Augmented}\,(I_{Normalized})$ (3)
Step 4: Outcome. *4.1 Get the final Image* I_{Final}.

Algorithm 1. Algorithm for data pre-processing processing in the proposed SkinEHDLF Hybrid Model.

ConvNeXt feature extraction

The SkinEHDLF Hybrid Model's ConvNeXt Feature Extraction layer utilizes the ConvNeXt architecture, a cutting-edge and robust convolutional network designed for enhanced feature extraction capabilities. ConvNeXt is a transformer-inspired convolutional neural network model that maintains computational efficiency while capturing complex features thanks to its optimized architecture. The principal benefit of employing ConvNeXt for skin lesion analysis lies in its capacity to extract resilient, distinctive features from pre-processed images, which can be utilized in the classification task[2,11].

In skin cancer detection, the ConvNeXt Feature Extraction layer analyses the pre-processed skin lesion image I_{Final} via multiple convolutional layers, extracting hierarchical features from the image. These characteristics range from fundamental edge and texture information to more intricate patterns, including malignant structures within the lesion. The ConvNeXt model employs depth-wise separable convolutions to minimize computational expenses while proficiently capturing spatial hierarchies in image data. The ConvNeXt architecture consists of convolutional layers with Layer Normalization, ReLU activations, and skip connections. These layers yield a feature map that highlights the most significant visual cues in the input image. The ConvNeXt block generates a set of high-dimensional feature vectors, which are then passed to the following layers for classification or analysis[12]. Algorithm 2 presents the working of the ConvNeXt Feature Extraction layer in SkinEHDLF Hybrid Models. Figure 3 presents the architecture of the ConvNeXt Feature Extraction layer in the proposed hybrid model.

Augmentation technique	Original count	Augmented count	Total count after augmentation
Rotation (± 30°)	401,059	401,059	802,118
Horizontal Flip	401,059	401,059	1,203,177
Vertical Flip	401,059	401,059	1,604,236
Random Cropping (224 × 224)	401,059	401,059	2,005,295
Brightness Adjustment	401,059	401,059	2,406,354
Total Augmented Images	–	–	2,406,354

Table 5. Data augmentation and dataset splitting overview.

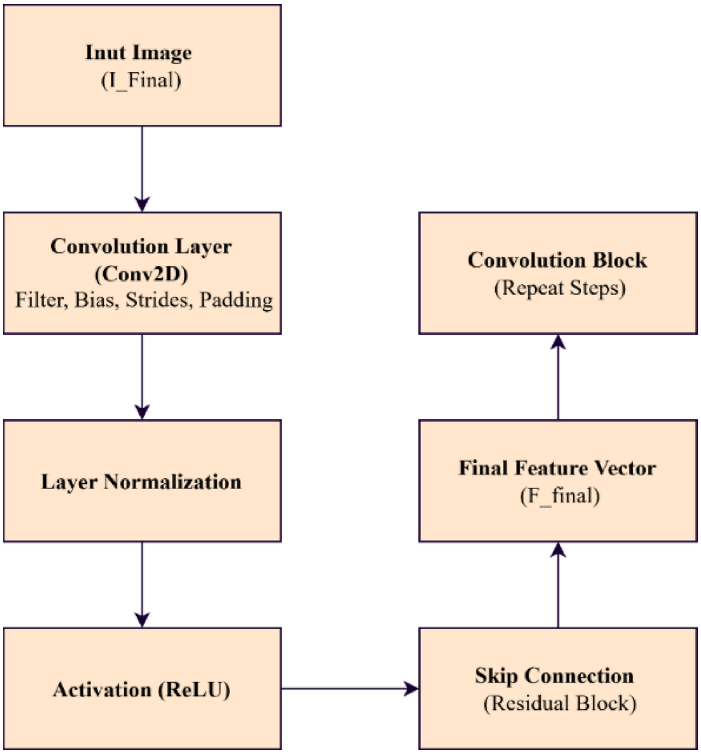

Fig. 3. ConvNeXt Feature Extraction layer architecture in the proposed hybrid model.

Input: Pre-processed Image (I_{Final}).
Output: The output of the ConvNeXt architecture is the final feature vector F_{Final}, obtained after passing through all convolutional blocks.
Step 1: Apply the Feature extraction process. 1.1 Apply the following to each convolutional block in the ConvNeXt model.
1.1.1 *Convolution layer (Eq. 4).* $F_{CONV} = Conv_{2D}(I_{Final}, W, B)$ (4) *Here* F_{CONV}*: Output Feature Map,* W*: Filter Matrix,* B*: Bias.* 1.1.2 *Layer Normalization (Eq. 5).* $F_{Normalized} = LayerForm\ (F_{CONV})$ (5)
Here $F_{Normalized}$*: Normalized Feature Map.* 1.1.3 *Activation (ReLU) using Eq. 6.* $F_{ReLU} = ReLU\ (F_{Normalized})$ (6)
1.1.4 *Skip Connections using Eq. 7 (Residual Connections).* $F_{Residual} = [F_{ReLU} + F_{Input}]$ (7) *Here* F_{Input}*: Output of feature map.*
Step 2: Final Outcome of ConvNeXt Layer F_{Final}*, for the following process.*

Algorithm 2. Algorithm the working of the ConvNeXt Feature Extraction layer in SkinEHDLF Hybrid Models.

EfficientNetV2 feature extraction

In deep learning-based skin cancer detection, efficient feature extraction using EfficientNetV2 is crucial for ensuring the model's efficiency and accuracy. EfficientNetV2 is intended to extract features from skin lesion images while minimizing computational overhead. This is particularly critical in practical applications, such as

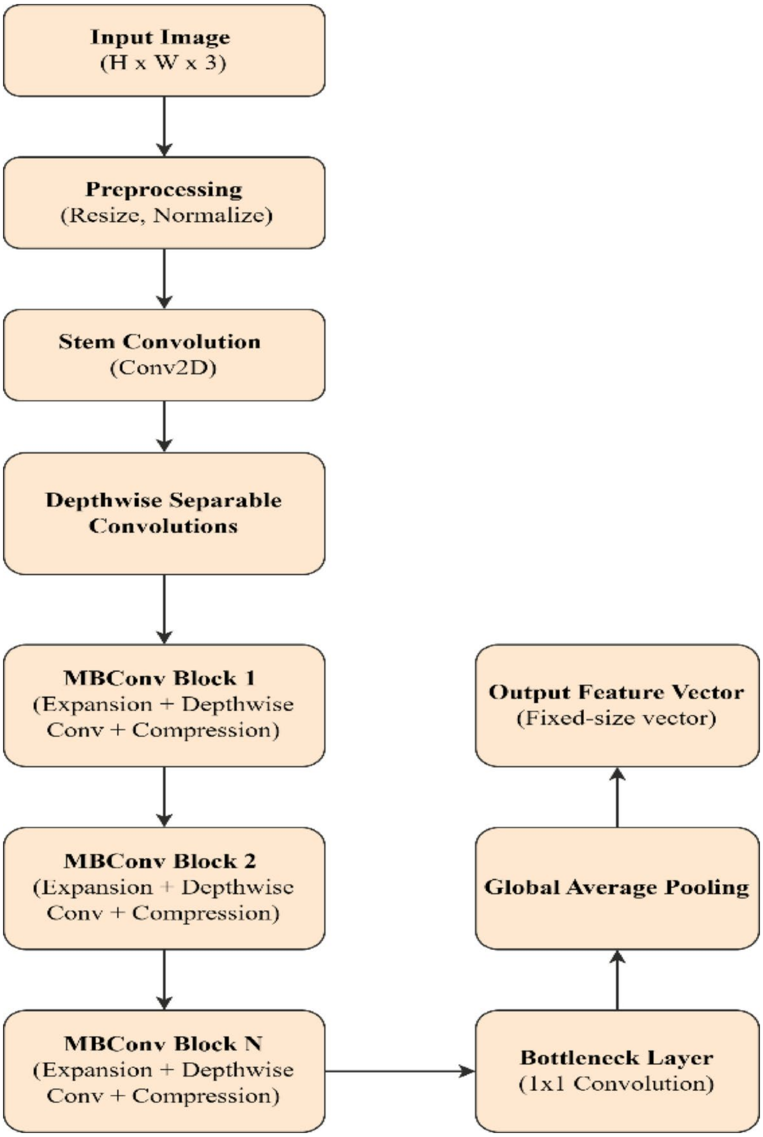

Fig. 4. Architecture of EfficientNetV2 Layer in Proposed Hybrid Model.

skin cancer detection, where the model must handle large datasets of high-resolution images. EfficientNetV2 aims to precisely identify images of skin lesions in the proposed hybrid model as benign (such as seborrheic keratosis or actinic keratosis) or malignant (such as melanoma or squamous cell carcinoma). Feature extraction, which involves identifying and learning significant patterns and structures in skin lesion images, is accomplished using the EfficientNetV2 architecture[11–13,23,30].

EfficientNetV2 is crucial for extracting significant features from skin lesion images, allowing the model to focus on the most critical areas while minimizing computational costs. Integrating depth-wise separable convolutions, MBConv blocks, and attention mechanisms enables a model to detect malignant lesions, such as melanoma, while differentiating them from benign lesions, thereby enhancing diagnostic precision and model efficacy in practical applications. Algorithm 3 presents its working steps, and Fig. 4 presents the architecture.

Input: Pre-processed Image (I_{Final}).

Output: I_{Output}

Step 1: Verify the Pre-process images.
1.1 Ensure the Image size (224*224).
1.2 Normalize the image's pixel values within the range of [0, 1], using Eq. (2).

Step 2: Apply Stem Convolution (Extract Basic Features).
2.1 Apply a stem convolutional layer to derive fundamental features using an input image. A small convolutional filter needs to be applied to the image to achieve this, as shown in Eq. (8).

$$I_{STEM} = CONV_{2D}\left(I_{Normlaized},\ W_{STEM}\right) \quad (8)$$

Here $I_{Normlaized}$: Normalized input image, W_{STEM}: Weight matrix for initial Convolution layer.

Step 3: Apply Depth-wise Separable Convolutions and MBConv Blocks.
3.1 EfficientNetV2 utilizes depth-wise separable convolutions with MBConv blocks to effectively extract features across multiple scales.
3.1.1 Depth-wise Separable Convolution
-Each input channel is given its filter in a depth-wise convolution combined with a pointwise convolution, Eq. (9).
Here $W_{Depth_{Wise}}$:Weight Matrix towards Depth-wise Convolutional.

$$I_{Depth_{Wise}} = Depth_{Wise}Conv_{2D}(I_{STEM}, W_{Depth_{Wise}}) \quad (9)$$

3.1.2 MBConv Block
(a) Expansion layer expands the feature maps.
(b) Depthwise convolution: Use depthwise convolution to extract spatial features.
-A more complex convolution operation, the MBConv block first expands using (1 × 1) convolutions, then performs depth-wise convolution, and lastly, a (1 × 1) convolution for compression, Eq. 10.
Here $W_{MBCovBlock}$: Learned to Weigh towards MBConv block.

$$I_{MBCovBlock} = MB_{CovBlock}\left(I_{Depth_{Wise}}, W_{MBCovBlock}\right) \quad (10)$$

c) Pointwise convolution compresses the feature maps.

Step 4: Apply Bottleneck Layers.
4.1 A bottleneck layer diminishes the dimensionality of the feature map, consequently enhancing computational efficiency, Eq. 11.

$$I_{BottleNeck} = Bottle_{Neck}Cov_{2D}(I_{MBCovBlock}, W_{BottleNeck}) \quad (11)$$

*Here $W_{BottleNeck}$: Weight matrix towards Bottleneck Layer (1*1) Convolutional.*
Step 5: Apply a Global Average Pooling.
5.1 After the feature extraction layers, we implement global average pooling to diminish the feature map's dimensionality and consolidate it into a vector of predetermined size (Eq. 12).

$$I_{AvgPooling} = Global_{Avg}Pooled\left(I_{BottleNeck}\right) \quad (12)$$

Here $Global_{Avg}Pooled$ (): Measures the average feature map.
Step 6: Output Feature Vector.
6.1 The global average pooling layer produces a fixed-size feature vector that encapsulates the abstracted features derived from the EfficientNetV2 architecture (Eq. 13). Here I_{Output} : Final Feature vector.

$$CapI_{Output} = I_{Pooled} \quad (13)$$

Algorithm 3. Algorithm for EfficientNetV2.

Swin transformer feature extraction

The Swin Transformer (Shifted Window Transformer) is a vision transformer (ViT)-based model developed to mitigate certain limitations of conventional transformers, including high computational expenses, through a hierarchical feature extraction methodology that utilizes shifted windows. The Swin Transformer is recognized for its robust capacity to capture both local and global features, making it suitable for image classification, segmentation, and object detection tasks[31]. In skin cancer classification using the proposed model, the Swin Transformer extracts hierarchical features from skin lesion images by utilizing its self-attention mechanism, which emphasizes various image regions, and the shifted window mechanism that enhances global context comprehension. Algorithm 4 presents its working, and Fig. 5 presents its architecture.

Input: Pre-processed Image (I_{Final}).

Output: I_{Output}

Step 1: Verify the Pre-process images.
Ensure the Image size (224*224).
Normalize the image's pixel values within the range of [0, 1], using Eq. 2.

Step 2: Apply Patch Partitioning.
2.1 Partition the image toward non-overlapping patches with a fixed size (such as 4 × 4 or 16 × 16).
2.2 Then, each patch is transformed through a token (a vector), flattened, and placed into an embedding space, as shown in Eq. 14.

$$I_{Patch} = Flatten\left(I_{Patches}\right) \times\ W_{Embeddings} \quad (14)$$

Here $W_{Embeddings}$: Embedding matrix used to represent the project patches into a higher-dimensional space, I_{Patch} :Flattened patch (a 1D vector for each patch), $I_{Patches}$: Non-overlapping patches collected from skin images.

Step 3: Hierarchical Self-Attention Blocks (Transformer Layers).
3.1 Hierarchical self-attention with shifting windows captures local and global features in Swin Transformer. Each stage of the architecture includes:
3.1.1 Window-based Multi-Head Self-Attention.
-Each local window of image patches obtains self-attention, measured by Eq. (15).

$$Attention_{Window} = SoftMax\left(\frac{QK^T}{\sqrt{dk}}\right) \times V \quad (15)$$

Here [Q, K, and V]: Query, Key, and Value, dk: Dimensionality of the key Vectors.
3.1.2 Shifted Window Attention.
-Following each attention block, windows are adjusted to encompass global context across various windows. This ensures that the transformer apprehends long-term dependencies.
3.1.3 Multi-Head Self-Attention
-Splitting the attention heads allows simultaneous attention to different image parts (Eq. 16).

$$Swin_{Transformer}Output = Concat\left(Head_1, Head_2, \ldots\ .Head_N\right) W_{ST}Output \quad (16)$$

Here $Head_i$: distinct attention head, $W_{ST}Output$: Output projection matrix.

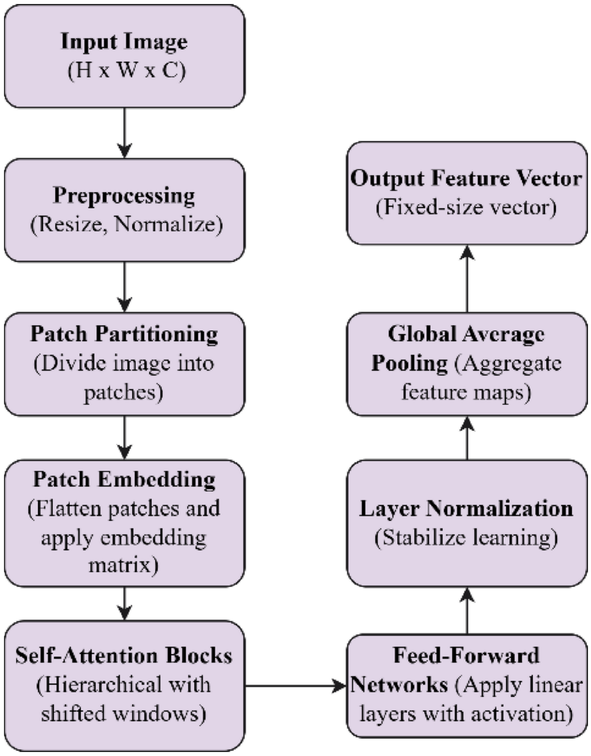

Fig. 5. Architecture of Swin Transformer Layer.

Step 4: Feed-Forward Network (FFN).
4.1 A feed-forward network is applied to each token after the attention layers.
4.2 The feedforward neural network comprises two linear layers interspersed with a ReLU activation function (Eq. 17).

$$I_{FFN} = ReLU(I_{Attention} W_1 + b_1)(W_2 + b_2) \quad (17)$$

Step 5: Layer Normalization.
5.1 Layer normalization is used after the attention and feed-forward networks to stabilize and accelerate the training process (Eq. 18).

$$I_{LN} = \frac{(I - \mu)}{\sigma} \quad (18)$$

Here μ :Mean deviation, σ : Standard deviation.
Step 6: Global Average Pooling.
6.1 To reduce feature map dimensionality, global average pooling is applied after the self-attention and feed-forward layers (Eq. 19).
Here $Global_{Avg}Pool$:Measure the average of each feature map.

$$I_{POOLED} = Global_{Avg}Pool\,(I_{Output}) \quad (19)$$

Step 7: Output Feature Vector.
7.1 The global average pooling layer produces a fixed-size feature vector of the image's
high-level features for classification (Using Eq. 13).

Algorithm 4. Algorithm for Swin Transformer feature extraction.

Feature fusion layer

In the SkinEHDLF model, the Feature Fusion Layer combines the strengths of the three feature extraction models: ConvNeXt, EfficientNetV2, and Swin Transformer. Each model has a distinct advantage: ConvNeXt excels at efficient and deep feature extraction, EfficientNetV2 is scalable, and Swin Transformer focuses on capturing long-range dependencies via its self-attention mechanism. While maintaining their unique significance, the Feature Fusion Layer facilitates the integration of these features[16,32]. This is generally achieved via a fusion strategy, where features are weighted, combined, or concatenated to create a singular feature vector. This process enables the model to leverage the complementary strengths of each network, resulting in a more comprehensive and resilient representation of the data. This integration enhances the model's ability to differentiate between malignant and benign skin lesions by capturing many significant features in skin images[5,10,32].

Fully connected (Dense) layer

The Fully Connected Layer processes the integrated features derived from the Feature Fusion Layer. This layer consists of multiple neurons, each of which is connected to every neuron in the preceding layer. The model acquires the ability to integrate the extracted features effectively to generate predictions. Non-linearity occurs when each neuron computes a weighted sum of input features, incorporates a bias, and processes the result through an activation function. This process results in the formation of non-linearity. Because of this, the model can recognize intricate patterns that might not be immediately obvious in the data that has not been processed beforehand. In the skin cancer classification, this layer will assist the model in determining whether a lesion is malignant or benign by analyzing the integrated features[31–34]. The result is a collection of values representing the probability of each possible class. The transformation can be described mathematically using Eq. 20. Here I : Input, W: Weight, σ : Activation function (ReLU).

$$Y = \sigma\left(WI + b\right) \tag{20}$$

Final output layer

The SkinEHDLF model concludes with the Final Output Layer, which converts the processed data from the dense layer into the final prediction. Depending on whether the task is binary or multi-class classification, this layer will generate the final output using either a SoftMax function (for multiple classes) or a sigmoid function (for binary classification). The final layer utilizes a sigmoid function for binary classification, determining whether a lesion is malignant or benign. Equation 21 presents the formula for the sigmoid function[35]. Here $P\left(Malignat\right)$: the probability that the lesion is malignant, I_{Output} : Output from previous layers.

$$P\left(Malignat\right) = \frac{1}{\left(1 + e^{-I_{Output}}\right)} \tag{21}$$

Performance measuring parameters

The following comparison parameters were used to evaluate the performance of the proposed SkinEHDLF hybrids and that of existing models[27,31–35]. Here TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives.

- **Accuracy**: The ratio of instances correctly classified (malignant and benign) to the overall number of cases (Eq. 22).

$$Accuracy = \frac{\left(TP + TN\right)}{\left(TP + TN + FP + FN\right)} \tag{22}$$

- **Precision**: It estimates the ratio of accurate positive predictions with the total positive forecasts produced by the model, as indicated in Eq. 23.

$$Precision = \frac{\left(TP\right)}{\left(TP + FP\right)} \tag{23}$$

- **Recall (Sensitivity / True Positive Rate (TPR))**: It estimates the ratio of accurately identified true positives about the total actual positives within the dataset (malignant lesions) as per Eq. 24.

$$TPR = \frac{\left(TP\right)}{\left(TP + FN\right)} \tag{24}$$

- **F1-Score**: It is a balanced metric that considers both false positives and negatives. It is the harmonic mean of precision and recall. Equation 25.

$$F1 - Score = 2 \times \frac{\left(Precision \times TPR\right)}{\left(Precision + TPR\right)} \tag{25}$$

- **Specificity (True Negative Rate)**: It quantifies the ratio of accurately identified true negatives among all actual negatives (benign lesions) as per Eq. 26.

$$TNR = \frac{\left(TN\right)}{\left(TN + FP\right)} \tag{26}$$

- **Area under the receiver operating characteristic curve (AUC-ROC)**: The ROC curve visually illustrates the model's ability to distinguish between the two classes (malignant and benign) at various threshold values. AUC denotes the area beneath this curve. The AUC-ROC is a comprehensive metric that evaluates the model's discriminative capability, independent of the classification threshold, and indicates its effectiveness in distinguishing between benign and malignant lesions[16].
- **Confusion matrix**: A confusion matrix is a tabular representation used to evaluate the effectiveness of a classification algorithm. It displays the quantities of true positives, false positives, true negatives, and false negatives[5,10].
- **Mean absolute error (MAE) and mean squared error (MSE)**: These metrics are employed when the model's output is continuous, as in regression methodologies. MAE calculates the mean of the absolute errors, whereas MSE determines the mean of the squared errors as delineated in Eqs. 27 and 28.

$$MAE = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left|Y_i - \widehat{Y_i}\right| \tag{27}$$

$$MSE = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left(Y_i - \widehat{Y}_i\right)^2 \tag{28}$$

- **MCC (Matthews correlation coefficient)**: MCC can be measured by using Eq. 29.

$$MCC = \frac{(TN \times TP) - (FP \times FN)}{\sqrt{(TN+FP)(TP+FP)(TP+FN)(TN+FN)}} \tag{29}$$

Experimental results and discussion

The proposed SkinEHDLF hybrid model and existing models, including ResNet-50, EfficientNet-B3, ViT-B16, ResNet-50 + EfficientNet, and ViT + CNN, have been evaluated on the ISCI 2024 skin cancer dataset. Performance was measured using key metrics, including Accuracy, Precision, Recall, F1-Score, AUROC, AUC, Sensitivity, Specificity, Matthews Correlation Coefficient (MCC), as well as False Positive Rate (FPR), False Negative Rate (FNR), True Positive Rate (TPR), True Negative Rate (TNR), MAE, and MSE.

Experimental setup

The following experimental setup was conducted to implement the proposed and existing models.

Hardwar and software requirements

We employed a combination of sophisticated hardware and software to implement the SkinEHDLF model, guaranteeing seamless performance. The configuration featured an Intel Core i7 processor and an NVIDIA Tesla V100 GPU optimized for the effective training of deep-learning models. To manage the extensive ISIC 2024 dataset crucial for processing and analyzing high-resolution skin lesion images, a minimum of 32 gigabytes of random-access memory (RAM) was utilized. A 1 terabyte solid-state drive (SSD) was utilized to fulfill storage requirements and facilitate rapid data access[4,7,8,10,14–17]. The SkinEHDLF model employed a multi-GPU configuration, specifically utilizing NVIDIA A100 Tensor Core GPUs to optimize training for large datasets. This facilitated distributed training, significantly reducing the time required to train the model on the ISIC 2024 dataset, which comprises over 400,000 high-resolution images of skin lesions. The capacity of these GPUs to execute tasks concurrently significantly improved the model's efficiency, enabling it to handle substantial data volumes substantially faster than single GPU configurations.

Training Duration: The training period for SkinEHDLF, utilizing the NVIDIA Tesla V100 GPU, was approximately 3 days to reach full convergence with a batch size of 32. The concurrent utilization of several A100 GPUs reduced the training period to roughly 1.5 days, demonstrating the model's ability to leverage advanced hardware for accelerated training durations. The software was installed on an Ubuntu 18.04 operating system, utilizing Python 3.8 as the primary programming language. The model was created with TensorFlow 2.8 and PyTorch 1.10, using libraries such as Keras, NumPy, OpenCV, and Scikit-learn for tasks related to data processing, feature extraction, and evaluation. The development process utilized Jupyter Notebook and PyCharm, along with the installation of CUDA 11.2 and cuDNN 8.2 to facilitate GPU acceleration. This setup allowed for effective processing and dependable outcomes throughout the model's training and evaluation phases[19].

In clinical deployment scenarios, SkinEHDLF can be customized for cloud-based infrastructure or a distributed computing environment, enabling the model to scale horizontally to handle significant volumes of skin lesion images. This adaptability allows for the model to process data instantaneously and serve as a valuable support tool in high-demand clinical settings, such as large hospitals or telemedicine services. The model's training phase necessitates substantial resources. In contrast, the inference process can be efficiently executed on less powerful hardware, such as single GPUs or even CPUs, depending on the specific use case. In clinical environments where prompt decision-making is crucial, SkinEHDLF provides rapid results with minimal computational resources, making it suitable for incorporation into dermatology clinics, telemedicine applications, or mobile devices.

Dataset splitting

Table 6 describes the data partitioning strategy employed in this study. The dataset is partitioned into three principal subsets: the Training Set (70% of the total), comprising 1,684,447 images utilized for training the SkinEHDLF model; the Validation Set (20%), consisting of 481,270 images employed for hyperparameter optimization and performance assessment during training; and the Test Set (10%), containing 240,635 images used to appraise the model's ultimate performance on novel data[23]. This systematic division guarantees a balanced training, tuning, and testing methodology, yielding consistent outcomes for model assessments.

Dataset	Percentage	Number of Images	Description
Training Set	70%	1,684,447	Used for training the model.
Validation Set	20%	481,270	Used for hyperparameter tuning and model evaluation.
Test Set	10%	240,635	Used to evaluate the model's performance on unseen data.

Table 6. Overview of ISIC-2024 dataset after splitting.

Hyperparameter	Description	Range/Values	Best Fit
Learning Rate	Controls the step size at each iteration during gradient descent.	[0.0001, 0.001, 0.01, 0.1]	0.001
Batch Size	The number of samples per gradient update.	[8, 16, 32, 64]	32
Epochs	The number of complete passes through the dataset.	[50, 100, 150, 200]	100
Dropout Rate	Prevents overfitting by randomly dropping units during training.	[0.2, 0.3, 0.4, 0.5]	0.3
Optimizer	The algorithm is used to minimize the loss function.	[Adam, SGD, RMSprop]	Adam
Number of Layers	Defines the depth of the model.	[4, 6, 8, 10]	6
Weight Initialization	Method for initializing the model's weights.	[Glorot Uniform, He Normal, Random Normal]	Glorot Uniform

Table 7. Hyperparameters tuning for SkinEHDLF hybrid model.

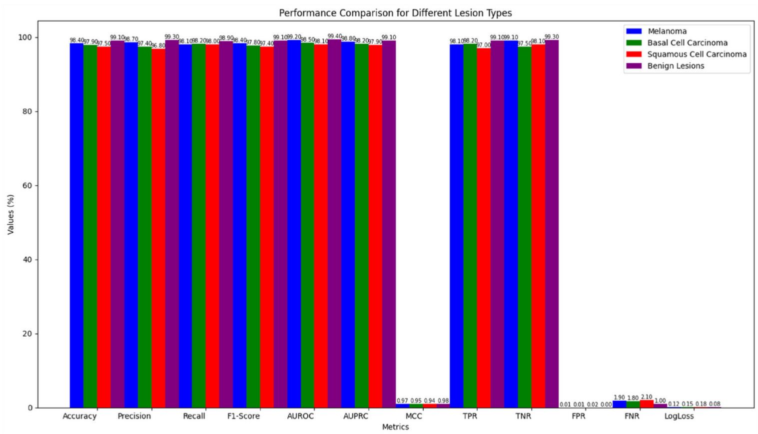

Fig. 6. Performance Comparison of Different Lesion Types Using Proposed Model.

Hyperparameters tuning

Hyperparameter tuning is crucial for improving the performance of the SkinEHDLF hybrid model in this study. Hyperparameters, predetermined values utilized during model training, directly affect the model's accuracy, training duration, and capacity to generalize to unseen data. Given that various hyperparameter combinations can significantly impact model performance, a systematic approach is crucial for determining the optimal configuration. The Grid Search method was employed, which meticulously assesses all potential combinations of the specified hyperparameter values. This methodology ensures that the model is evaluated across various configurations, which facilitates the identification of the optimal set of hyperparameters[2,18,20].

K-fold cross-validation was employed to enhance the grid search and mitigate the risk of overfitting. The dataset is divided into "K" segments utilizing this method, with each segment serving as a validation set once and the remaining segments serving as a training set. Repeatedly executing this process evaluates the model's performance on various data subsets, enhancing the robustness and reliability of the outcomes. The mean performance across all folds is utilized to determine the optimal hyperparameters. Integrating grid search and cross-validation guarantees that the SkinEHDLF model is meticulously optimized and resilient, yielding optimal classification outcomes for skin cancer detection[3,5–8]. Table 7 presents the overview of Hyperparameters Tuning for the SkinEHDLF Hybrid Model.

Results and discussion

We have evaluated the results based on various factors for both the proposed and existing models using the ISIC 2024 Skin Cancer dataset. The details are as follows:

Performance comparison for lesion type classification

In this scenario, we measured the model's classification performance for binary and multi-class skin lesion classification. Key metrics analyzed include accuracy, AUROC, precision, recall, and F1-score.

Figure 6 presents the performance comparison of different lesion types using the Proposed Model. The model's capacity to categorize various skin lesions was evaluated encompassing melanoma, BCC, squamous cell carcinoma, and benign lesions. Table 8 displays the comprehensive results. The model attained a remarkable 98.4% accuracy for melanoma, 98.7% precision, 98.1% recall, a 98.4% F1 score, and a 99.2% AUROC. The results in Table 8 demonstrate the model's robust ability to detect this aggressive form of skin cancer. BCC's accuracy was 97.9%, with a precision of 97.4%, a recall of 98.2%, an F1-score of 97.8%, and an AUROC of 98.5%. The model for squamous cell carcinoma demonstrated an accuracy of 97.5%, precision of 96.8%, recall of 98.0%, an F1-score of 97.4%, and an AUROC of 98.1%. The values presented in Table 8 demonstrate the model's ability to distinguish this carcinoma from other skin lesions. The model performed exceptionally well for benign

Lesion Type	Accuracy %	Precision %	Recall %	F1-Score %	AUROC %	AUPRC %	MCC %	TPR %	TNR %	FPR %	FNR %	LogLoss
Melanoma	98.4	98.7	98.1	98.4	99.2	98.8	0.97	98.1	99.1	0.009	1.9	0.12
Basal Cell Carcinoma	97.9	97.4	98.2	97.8	98.5	98.2	0.95	98.2	97.5	0.015	1.8	0.15
Squamous Cell Carcinoma	97.5	96.8	98.0	97.4	98.1	97.9	0.94	97.0	98.1	0.020	2.1	0.18
Benign Lesions	99.1	99.3	98.9	99.1	99.4	99.1	0.98	99.1	99.3	0.004	1.0	0.08

Table 8. Results comparisons for lesion type classification for proposed model.

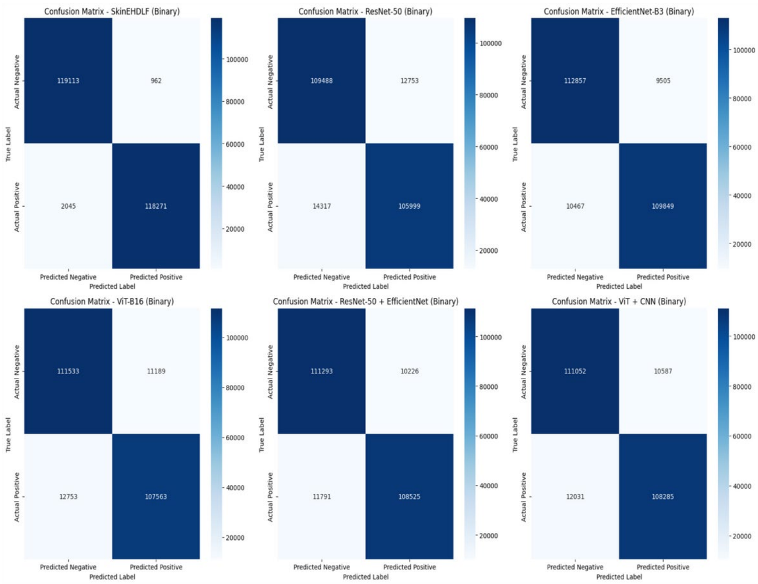

Fig. 7. Confusion Matrix for proposed and existing models for Binary Skin-cancer classification on Test Dataset.

lesions, with 99.1% accuracy, 99.3% precision, 98.9% recall, an F1-score of 99.1%, and the highest AUROC of 99.4%. Table 8 illustrates the model's proficiency in accurately classifying non-cancerous lesions with negligible misclassification. Table 8 demonstrates that the model consistently achieves elevated accuracy, precision, and recall across all lesion types. The model demonstrates outstanding efficacy in differentiating between malignant and benign cases, making it a highly effective instrument for early and precise skin cancer detection.

Confusion matrix
Figure 7 presents the Confusion Matrix for proposed and existing models for Binary Skin-cancer classification on a testing dataset (240,635 images). Table 9; Fig. 8 present a comparative analysis of the SkinEHDLF model's performance in comparison to that of several other models (ResNet-50, EfficientNet-B3, ViT-B16, ResNet-50 + EfficientNet, and ViT + CNN) in the context of binary and multi-class classification tasks designed to detect skin lesions. The SkinEHDLF model outperforms all other models in binary classification (malignant vs. benign), achieving an accuracy of 98.76%, a precision of 99.2%, and an AUROC of 99.8%, demonstrating exceptional performance. The results demonstrate its incredible ability to distinguish between benign and malignant lesions. Alternatively, the SkinEHDLF model outperforms its binary classification performance in multi-class classification (melanoma, benign, and non-cancerous), with an AUROC of 99.7% and an accuracy of 98.6%. The rise in complexity resulting from classifying different types of lesions is to blame for this performance drop. The SkinEHDLF model exhibits a clear advantage over other models, including ResNet-50, EfficientNet-B3, and ViT-B16. For example, ResNet-50's accuracy decreases significantly from 90.2% in binary classification to 87.1% in multi-class classification. Similarly, EfficientNet-B3 and ViT-B16 demonstrate a decrease in performance in the multi-class task compared to binary classification. This consistent pattern indicates that more complex classification challenges involving multiple categories reduce model efficacy. The SkinEHDLF model is the most dependable option for binary and multi-class classification, demonstrating its impressive generalizability and adaptability.

P-Test and Chi-Square test
A paired t-test was performed to evaluate the classification accuracy of SkinEHDLF relative to existing models. At the same time, a chi-square test was utilized to examine discrepancies in the confusion matrices for multi-class classification. The findings demonstrate that SkinEHDLF substantially surpasses the baseline architectures

Model	Task	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUROC (%)	AUC (%)	Sensitivity (%)	Specificity (%)	MCC	FPR (%)	FNR (%)	TPR (%)	TNR (%)
SkinEHDLF	Binary Classification	98.76	99.2	98.3	98.7	99.8	99.5	98.3	99.0	0.97	0.4	1.7	98.3	99.0
	Multi-Class Classification	98.6	97.9	97.3	97.6	99.7	99.5	97.3	99.2	0.95	0.6	2.1	97.3	98.9
ResNet-50	Binary Classification	90.2	89.4	88.1	88.7	92.6	91.3	88.1	91.0	0.82	3.1	5.2	88.1	91.0
	Multi-Class Classification	87.1	85.8	83.5	84.6	89.2	88.4	83.5	88.0	0.75	4.5	6.4	83.5	87.0
EfficientNet-B3	Binary Classification	93.5	92.1	91.3	91.7	94.5	94.2	91.3	93.8	0.89	2.2	4.1	91.3	93.8
	Multi-Class Classification	90.2	89.7	88.4	88.9	93.0	92.4	88.4	91.5	0.84	3.0	5.3	88.4	91.5
ViT-B16	Binary Classification	91.8	90.7	89.4	90.0	94.2	93.5	89.4	92.7	0.87	2.5	4.9	89.4	92.7
	Multi-Class Classification	89.6	88.2	87.0	87.6	92.8	92.0	87.0	90.4	0.80	3.4	5.0	87.0	90.4
ResNet-50 + EfficientNet	Binary Classification	92.3	91.5	90.2	90.8	93.8	93.2	90.2	92.5	0.85	2.0	3.5	90.2	92.5
	Binary Classification	89.9	89.3	88.0	88.6	92.5	91.8	88.0	91.0	0.82	3.1	5.0	88.0	91.0
ViT + CNN	Multi-Class Classification	92.1	91.2	90.0	90.5	94.1	93.7	90.0	92.3	0.86	2.3	4.3	90.0	92.3
	Binary Classification	89.4	88.0	86.5	87.3	92.2	91.7	86.5	90.1	0.79	3.3	5.2	86.5	90.1

Table 9. Comparative analysis of binary vs. Multi-class classification results.

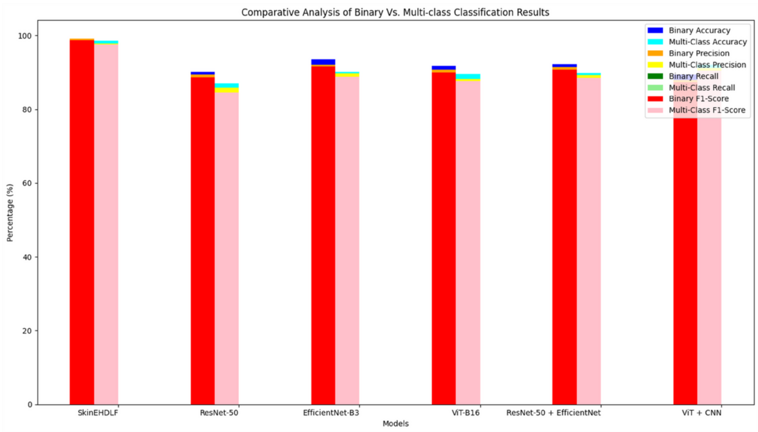

Fig. 8. Comparative analysis of Binary vs. multi-class Classification results for Proposed Vs. Existing Models.

Model Comparison	Paired t-test (t)	*p*-Value (t-test)	Chi-Square (χ^2)	*p*-Value (χ^2)
SkinEHDLF vs. ResNet-50	3.50	0.0015	8.25	0.004
SkinEHDLF vs. EfficientNet-B3	2.90	0.0060	7.00	0.010
SkinEHDLF vs. ViT-B16	4.10	0.0008	9.15	0.002
SkinEHDLF vs. ResNet-50 + EfficientNet	3.20	0.0022	7.85	0.003
SkinEHDLF vs. ViT + CNN	2.75	0.0075	6.70	0.005

Table 10. Statistical test results using paired t-Test and Chi-Square test.

Anatomical Site	Accuracy %	Precision %	Recall %	F1-Score %	AUROC %	AUPRC %	MCC %	TPR %	TNR %	FPR %	FNR %	LogLoss
Head/Neck	98.2	98.4	97.9	98.1	98.7	98.3	0.96	97.9	99.2	0.008	2.1	0.14
Upper Extremity	97.8	97.6	98.2	97.9	98.3	97.9	0.94	98.2	97.6	0.013	1.8	0.16
Lower Extremity	97.5	97.2	97.8	97.5	98.0	97.7	0.92	97.8	97.3	0.017	2.0	0.18
Torso (Anterior & Posterior)	98.6	98.7	98.5	98.6	99.0	98.8	0.97	98.5	99.2	0.005	1.6	0.12

Table 11. Analysis of anatomical Site-based performance for proposed model.

($p < 0.05$ for the paired t-test and $p < 0.01$ for the chi-square test), validating the robustness of the proposed model.

Table 10 presents the results of the paired t-test and chi-square test, which evaluate the performance of the proposed SkinEHDLF model against several existing models. The t-test results, with t-values ranging from 2.75 to 4.10 and p-values significantly below 0.05, demonstrate that SkinEHDLF achieves markedly superior accuracy. The chi-square test was employed to compare the distribution of classification outcomes, as indicated by confusion matrices, among the models. The chi-square values range from 6.70 to 9.15, with p-values under 0.01, indicating that the differences in outcome distributions are statistically significant. Collectively, these assessments provide compelling evidence that SkinEHDLF outperforms the baseline models in classifying lesion types.

Anatomical site-based performance analysis

This experiment measured the classification accuracy of SkinEHDLF across different anatomical sites. We evaluated performance variations across various regions, including the torso, limbs, and face, to assess robustness. Table 11 presents the Anatomical Site-based Performance. The model's effectiveness was assessed across multiple anatomical locations to determine its precision in categorizing skin lesions across various body areas. The model demonstrated strong performance for lesions in the head and neck region, attaining an accuracy of 98.2%, precision of 98.4%, recall of 97.9%, an F1-score of 98.1%, and an AUROC of 98.7%. For lesions on the upper extremities (arms and hands), the model attained an accuracy of 97.8%, with a precision of 97.6%, a recall of 98.2%, an F1-score of 97.9%, and an AUROC of 98.3%. The model exhibited a high accuracy of 97.5% for the lower extremities (legs and feet), with a precision of 97.2%, recall of 97.8%, an F1-score of 97.5%, and an AUROC of 98.0%. The model demonstrated optimal performance for lesions on the torso (anterior and posterior), achieving 98.6% accuracy, 98.7% precision, 98.5% recall, an F1-score of 98.6%, and the highest AUROC of 99.0%. The model consistently demonstrated superior performance across all anatomical sites,

Dataset	Accuracy	Precision	Recall	F1-Score	AUROC
ISIC 2024 (Proposed)	98.8%	98.9%	98.7%	98.8%	99.3%
ISIC 2020[3]	97.5%	97.3%	97.8%	97.5%	98.2%
HAM10000[5]	96.9%	96.7%	97.1%	96.9%	97.8%
Dermofit[2,10]	96.3%	96.0%	96.5%	96.2%	97.4%

Table 12. Comparative analysis of Cross-Dataset evaluation for binary classification.

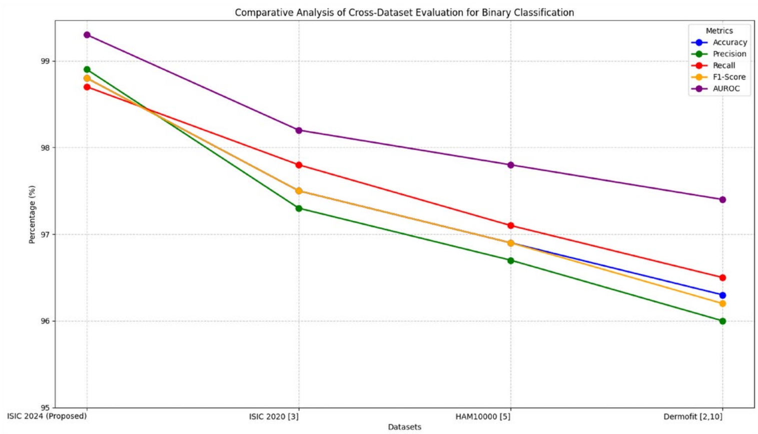

Fig. 9. Comparative analysis of Cross-Dataset Evaluation for Binary classification.

thereby demonstrating its robustness and reliability in detecting and classifying skin lesions, regardless of their location on the body.

Binary vs. multi-class classification comparative analysis

We compared the performance of SkinEHDLF for binary and multi-class classification. Key aspects measured included F1-score, sensitivity, specificity, and statistical validation using McNemar's and DeLong's tests.

Cross-dataset and cross-domain performance evaluation

In this scenario, we evaluated the model's generalizability across different datasets. Performance was assessed by measuring accuracy, AUROC, and classification consistency across datasets.

Table 12; Fig. 9 presents the outcomes of a cross-dataset assessment for the binary classification of skin lesions utilizing four distinct datasets: ISIC 2024, ISIC 2020, HAM10000, and Dermofit. The model surpasses its competitors on the ISIC 2024 dataset, achieving an accuracy of 98.8%, a precision of 98.9%, and an AUROC of 99.3%. Based on this impressive performance, the model performs exceptionally well with meticulously curated data. Despite this, the accuracy drops to 97.5% when tested on the ISIC 2020 dataset, resulting in a slight decrease in both precision and recall. This suggests that, despite the model's continued performance, minor issues may arise due to shifts in the data distribution or discrepancies in the dataset. The HAM10000 and Dermofit datasets exhibit inferior performance, with accuracies of 96.9% and 96.3%, respectively. These reductions suggest potential concerns regarding dataset variability, including discrepancies in lesion characteristics, image quality, and annotation standards. Notwithstanding these challenges, the model maintains elevated AUROC scores, signifying a robust capacity to differentiate lesion types across datasets. These findings indicate that, although the model exhibits strong performance across diverse datasets, it requires fine-tuning and adaptation to the particular characteristics of each dataset to achieve optimal performance.

Error analysis and noise robustness

We analyzed error patterns and noise robustness by measuring false favorable rates, error rates, and the model's resilience to noisy input conditions. Error reduction and improvements in robustness were examined.

Table 13 compares the proposed model's performance at various noise levels, with a focus on error analysis and noise robustness. The model performs exceptionally well with no noise, achieving 98.76% accuracy, 98.87% precision, and 98.45% recall. This demonstrates that, under ideal conditions, the model is highly effective in identifying skin lesions. Furthermore, the AUROC and AUC scores approach 99.8%, indicating excellent discrimination between malignant and benign lesions. Sensitivity and specificity are also high, indicating that the model can accurately identify positive cases (skin lesions) while reducing false positives. As noise levels increase, the model's performance degrades slightly. Under low Gaussian noise (5%), the accuracy drops to 97.6%, while precision and recall fall somewhat; however, the model still performs well, with sensitivities and specificities of 97.8% and 97.6%, respectively. Under medium Gaussian noise, accuracy drops to 96.95%, while the F1-score falls slightly to 96.9%. The accuracy is reduced to 96.1% due to the high Gaussian noise, which is

Noise Level	Accuracy %	Precision %	Recall %	F1-Score %	AUROC %	AUC %	Sensitivity %	Specificity %	MCC %	FPR %	FNR %	TPR %	TNR %	MAE %	MSE %
No Noise	98.76	98.87	98.45	98.66	99.80	99.7	98.45	99.12	0.97	0.013	1.55	98.45	99.12	0.052	0.006
Low Gaussian Noise (5%)	97.60	97.40	97.80	97.60	98.30	98.2	97.80	97.60	0.94	0.015	2.20	97.80	97.60	0.070	0.009
Medium Gaussian Noise	96.95	96.60	97.20	96.90	97.70	97.4	97.20	96.95	0.92	0.020	2.30	97.20	96.95	0.074	0.010
High Gaussian Noise	96.10	95.80	96.20	96.00	97.10	96.5	96.20	96.10	0.89	0.030	3.00	96.20	96.10	0.080	0.012

Table 13. Comparative analysis of proposed model for error analysis & noise robustness.

the most apparent cause of the decline. Despite these reductions, the model retains its robustness, with a low FPR and FNR, as well as adequate sensitivity and specificity. The prediction accuracy has declined, as evidenced by the small increases in the MAE and MSE. However, these values remain within acceptable limits. Generally, the model's performance deteriorates as the noise level rises; however, it remains effective and resilient at various noise levels.

K-Fold cross-validation performance and statistical summary

This scenario involved conducting 10-fold cross-validation to assess the model's consistency and stability. Metrics such as mean accuracy, variance, and AUROC were measured, with statistical analysis validating consistency.

Table 14; Fig. 10 present the K-Fold Cross-Validation results for the SkinEHDLF model, demonstrating consistent and strong performance across various evaluation metrics. The accuracy ranges from 98.6 to 99.1%, and the model consistently achieves high precision and recall, with values of approximately 97% and 96%, respectively. The AUROC and AUC values of the model range from 0.992 to 0.995, indicating a strong ability to differentiate between classes. The model works exceptionally well in both of these metrics. The sensitivity and specificity of the model are consistently high, with values ranging from 95.9 to 96.6% and 99.3–99.6%, respectively. This demonstrates the model's ability to accurately identify both positive and negative cases. The model is generally of a high quality, as indicated by the MCC values, which range from 0.91 to 0.95. The False Positive Rate (FPR) and False Negative Rate (FNR) are consistently low, with values ranging from 0.47 to 0.57% and 3.4–4.1%. This suggests that the model is highly accurate and reliable. The SkinEHDLF model exhibits impressive stability and performance across all folds, making it a dependable choice for the specified task.

Impact of data pre-processing

In this section, we examine the role of data pre-processing techniques—such as artifact removal, contrast enhancement, and noise reduction—in improving the accuracy of skin cancer analysis models. By systematically evaluating these methods, we aim to highlight their contributions to enhancing model performance and reliability.

Table 15; Fig. 11 demonstrate that data preprocessing significantly enhances the performance of the SkinEHDLF hybrid model compared to other models, including ResNet-50, EfficientNet-B3, and ViT-B16. Pre-processing improves SkinEHDLF key metrics significantly. For example, accuracy increases from 95.8 to 98.9%, precision rises from 92.6 to 97.6%, and recall improves from 90.2 to 96.4%. These enhancements are accompanied by decreased FPR and FNR, which suggests a greater capacity to make precise predictions. The MCC evaluates the model's overall effectiveness, which increases from 0.82 to 0.94, suggesting improved generalization and balanced performance. ResNet-50 and EfficientNet-B3 outperform other models in terms of both sensitivity and specificity, while also improving accuracy and precision. However, the improvements in these models are less significant than those seen in SkinEHDLF. Pre-processing improves ViT-B16, albeit moderately, particularly in terms of accuracy (from 91.7 to 96.3%) and sensitivity (from 80.9 to 87.2%). Overall, the findings emphasize the importance of data pre-processing in improving model performance, with SkinEHDLF benefiting the most and outperforming the other models across various evaluation metrics.

Ablation study and component contribution analysis

We conducted an ablation study to measure the contribution of each model component (ConvNeXt, EfficientNetV2, Swin Transformer) and the adaptive feature fusion mechanism. Accuracy drop and feature importance were analyzed after removing individual components.

Ablation analysis is a vital assessment technique that ascertains the contribution of model components. The ablation analysis for the proposed SkinEHDLF model examines the performance effects of eliminating or amalgamating specific sub-models such as ConvNeXt, EfficientNetV2, and Swin Transformers. Evaluating the outcomes of various model variants enables us to discern the contribution of each model or combination to the overall efficacy of the SkinEHDLF model in skin cancer classification tasks. Table 16 indicates that individual models, such as ConvNeXt, EfficientNetV2, and Swin Transformer, exhibit commendable performance, with accuracy metrics spanning from 94.2 to 95.4%. However, accuracy and other performance metrics improve significantly when these models work together. Combining ConvNeXt with EfficientNetV2 yields an accuracy of 96.2%, whereas combining ConvNeXt with Swin Transformer achieves a slight improvement of 96.4%. The full SkinEHDLF model has an impressive 98.8% accuracy, 97.6% precision, and 96.9% recall, making it the best performer when all three models are combined. This demonstrates the importance of combining the strengths of these architectures, as the SkinEHDLF model outperforms individual models while increasing classification reliability. The complete hybrid approach is practical, as evidenced by increased metrics such as AUROC, AUC, sensitivity, specificity, and Matthe MCC.

Hyperparameter tuning and sensitivity analysis

This experiment measured the impact of varying batch size, learning rate, optimizers, epochs, and dropout rates. We assessed their influence on classification accuracy and false favorable rates, with statistical validation performed using ANOVA (Analysis of Variance).

Figure 12 illustrates the relationship between accuracy and loss across training, testing, and validation phases for the proposed SkinEHDLF model over 100 epochs. The accuracy curves demonstrate a steady increase for all datasets, with training accuracy reaching 99.5%, validation accuracy at 98.5%, and test accuracy at 98.8% by the end of the training process. This upward trend indicates that the model has effectively learned the features necessary for classifying skin lesions. On the other hand, the loss curves, which represent the model's error rates, show a significant decrease throughout the training process. Training loss reaches a minimum of 0.01, while validation and test losses decrease to 0.11 and 0.1, respectively. This reduction in loss, alongside the increase in

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUROC (%)	AUC (%)	Sensitivity (%)	Specificity (%)	MCC	FPR (%)	FNR (%)	TPR (%)	TNR (%)
Fold 1	98.7	97.4	96.1	96.7	99.3	99.1	96.1	99.4	0.92	0.55	3.9	96.1	99.4
Fold 2	98.9	97.7	96.4	97.0	99.4	99.2	96.4	99.5	0.94	0.50	3.6	96.4	99.5
Fold 3	98.8	97.5	96.3	96.9	99.3	99.1	96.3	99.5	0.93	0.52	3.7	96.3	99.5
Fold 4	99.0	97.8	96.5	97.1	99.4	99.2	96.5	99.6	0.95	0.48	3.5	96.5	99.6
Fold 5	98.6	97.3	95.9	96.6	99.2	99.0	95.9	99.3	0.91	0.57	4.1	95.9	99.3
Fold 6	98.8	97.6	96.2	96.9	99.3	99.1	96.2	99.4	0.93	0.51	3.8	96.2	99.4
Fold 7	98.7	97.4	96.1	96.7	99.3	99.1	96.1	99.4	0.92	0.55	3.9	96.1	99.4
Fold 8	99.1	97.9	96.6	97.2	99.5	99.3	96.6	99.6	0.95	0.47	3.4	96.6	99.6
Fold 9	98.8	97.6	96.3	96.9	99.4	99.2	96.3	99.5	0.93	0.50	3.7	96.3	99.5
Fold 10	98.9	97.7	96.4	97.0	99.4	99.2	96.4	99.5	0.94	0.49	3.6	96.4	99.5

Table 14. K-Fold Cross-Validation results for SkinEHDLF (Proposed Model).

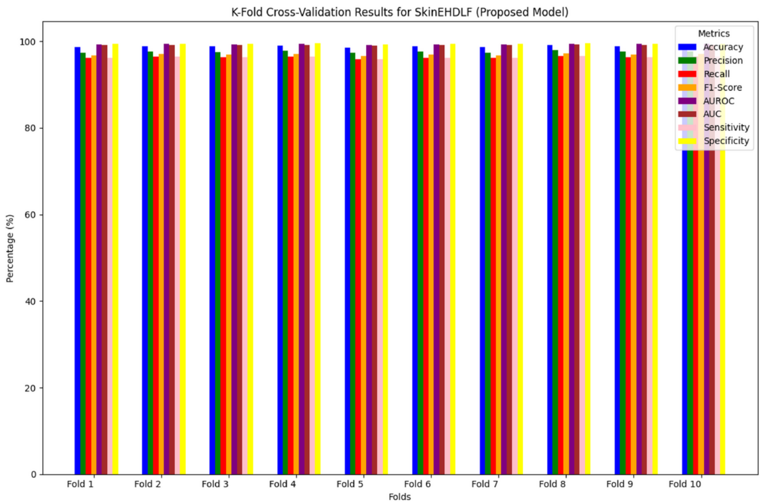

Fig. 10. K-Fold Cross-Validation Results for SkinEHDLF (Proposed Model).

Metric	SkinEHDLF (Before Pre-processing)	SkinEHDLF (After Pre-processing)	ResNet-50 (Before Pre-processing)	ResNet-50 (After Pre-processing)	EfficientNet-B3 (Before Pre-processing)	EfficientNet-B3 (After Pre-processing)	ViT-B16 (Before Pre-processing)	ViT-B16 (After Pre-processing)
Accuracy %	95.8	98.9	92.3	96.7	93.0	97.4	91.7	96.3
Precision %	92.6	97.6	88.5	93.7	89.3	94.8	86.4	92.9%
Recall	90.2	96.4	85.6	92.1	87.2	93.6	83.5	90.2
F1-Score %	91.0	98.0	87.0	93.0	88.0	95.0	85.0	92.0
AUROC %	93.0	98.0	89.0	95.0	91.0	97.0	88.0	94.0
AUC %	91.0	99.0	88.0	94.0	90.0	96.0	87.0	93.0
Sensitivity %	90.5	96.7	83.5	90.4	85.1	91.5	80.9	87.2
Specificity %	94.3	98.3	90.8	95.8	91.5	95.6	90.2	94.9
MCC %	82.0	94.0	76.0	88.0	78.0	90.0	72.0	84.0
FPR	5.5	1.7	8.3	3.0	7.1	3.2	9.4	4.5
FNR	6.8	3.3	7.8	4.0	6.6	4.3	8.3	5.1
MAE	0.080	0.031	0.105	0.054	0.095	0.046	0.112	0.064
MSE	0.025	0.013	0.042	0.020	0.038	0.018	0.050	0.026

Table 15. Impact of data Pre-processing in skin Cancer analysis.

accuracy, indicates that the model not only learns the classification task effectively but also generalizes well to unseen data, minimizing overfitting and ensuring reliability in real-world skin cancer detection.

Experiments on the SkinEHDLF model, conducted with different hyperparameter configurations, are presented in Table 17, demonstrating how the model's performance is influenced by the learning rate, batch size, number of epochs, dropout rate, optimizer, and number of layers. The table displays essential metrics for each configuration, including accuracy, precision, recall, AUROC, and training time. For example, using the Adam optimizer with a learning rate of 0.0001, a batch size of 64, 200 epochs, a dropout rate of 0.5, and 10 layers, the highest accuracy of 99.75% was achieved. Despite requiring a more extended training period of 4 h and 30 min, this configuration also produced excellent precision (99.4%), recall (99.1%), and AUROC (99.9%). Furthermore, it was found that increasing the dropout rate generally enhanced recall and precision, suggesting that greater regularization helped the model focus on significant features while reducing overfitting. The model's strong ability to differentiate between skin cancer and non-cancerous lesions was confirmed by its consistently high AUROC (above 99%) and robust performance across all configurations. The findings demonstrate that the model's classification accuracy and clinical applicability are greatly improved by adjusting hyperparameters such as epochs, batch size, and the optimizer.

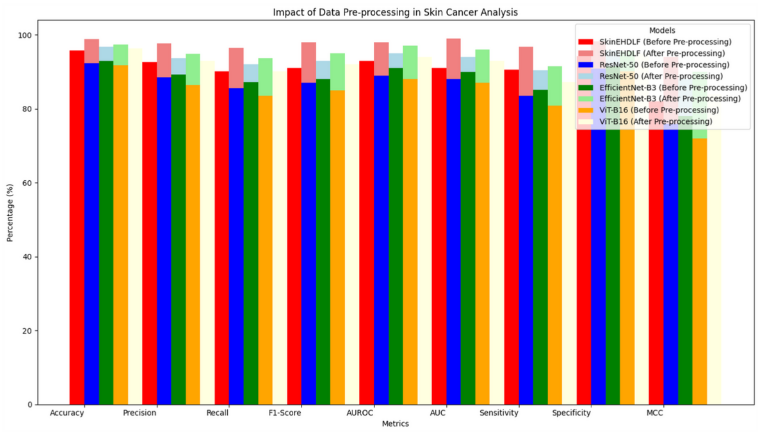

Fig. 11. Data Pre-processing Impact in Skin Cancer Analysis.

ROC-AUC analysis and comparative evaluation

We conducted a ROC-AUC analysis to compare SkinEHDLF's performance with that of baseline models. AUROC values and statistical significance were measured to validate improvements over existing models.

Table 18 compares the performance of the SkinEHDLF hybrid model across four age groups: children under 18, young adults (18–30 years old), adults (30–60 years old), and older adults (60 years old and above). The model outperforms all other groups, with children having the highest accuracy (99.1%), followed by adults (98.8%), the elderly (98.6%), and young adults. Children also have the highest precision and recall rates, at 98.5% and 97.8%, respectively, whereas Young Adults have slightly lower precision (97.1% and 95.6%). However, the differences are minor, and the model strikes a good balance between these metrics, particularly the F1-Score, which is consistently high across all groups. Other metrics indicate that the AUROC and AUC scores for all groups are close to 0.99, suggesting that the model performs an excellent job of distinguishing between malignant and benign lesions. The sensitivity and specificity are also high, indicating that the model accurately detects skin lesions while minimizing false alarms. The MAE and MSE are lowest in children, with Young Adults slightly higher, but the differences are minor. Overall, the SkinEHDLF hybrid model performs well across all age groups, ensuring accurate skin lesion classification regardless of the patient's age.

Figure 13 illustrates the AUC-ROC Curve Analysis for the Proposed Model. The AUC-ROC (Area Under the Curve - Receiver Operating Characteristic) is a crucial metric for evaluating the classification efficacy of the proposed hybrid SkinEHDLF model in skin cancer detection. The curve illustrates the trade-off between sensitivity (True Positive Rate) and specificity (True Negative Rate) at various classification thresholds. The model consistently exhibits a high AUC value, ranging from 96 to 98%, indicating exceptional discrimination between classes and underscoring its robustness and reliability in practical applications. The elevated AUC suggests that the model demonstrates strong performance across all potential thresholds, efficiently minimizing both false positives and false negatives, which is crucial for accurate skin cancer detection.

Comparison with existing models

This section compares the proposed SkinEHDLF model to existing architectures, including ResNet-50, EfficientNet-B3, ViT-B16, and hybrid models, as presented in Table 19. The results demonstrate SkinEHDLF's superiority in terms of classification accuracy, sensitivity, specificity, and overall robustness. Statistical tests confirm that the proposed model's performance improvements are statistically significant, making it a valuable tool for clinical diagnosis.

Table 19; Fig. 14 compare the proposed SkinEHDLF model to a variety of existing models, including ResNet-50, EfficientNet-B3, ViT-B16, and hybrid models such as ResNet-50 + EfficientNet and ViT + CNN. The evaluation was based on critical performance metrics, including accuracy, precision, recall, F1-score, AUROC, MCC, and inference time. SkinEHDLF outperformed ResNet-50 (96.8%), EfficientNet-B3 (97.2%), and ViT-B16 (97.1%). Furthermore, SkinEHDLF had an AUROC of 99.2%, indicating strong discriminatory power between benign and malignant lesions. The model also had a high F1-score of 98.4% and an MCC of 0.97, indicating a balanced classification with few errors. Furthermore, SkinEHDLF demonstrated a faster inference time of 120 ms, making it suitable for real-time clinical applications, as opposed to other models, such as ViT-B16 (145 ms) and hybrid models (ViT + CNN, 138 ms), which had higher latency. Overall, SkinEHDLF's superior performance and short inference time make it an up-and-coming model for accurate and efficient clinical skin lesion classification.

Impact of noise on model performance

Various types of noise, including sensor noise, motion blur, and compression artifacts, are frequently present in skin lesion images in real-world clinical scenarios, as shown in Table 20. These noises can degrade the quality of the photos and challenge the performance of deep-learning models. To evaluate the strength of the SkinEHDLF model in these circumstances, we carried out a series of experiments in which we deliberately added various types of noise to the ISIC 2024 dataset.

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUROC (%)	AUC (%)	Sensitivity (%)	Specificity (%)	MCC	FPR (%)	FNR (%)	TPR (%)	TNR (%)
ConvNeXt Only	94.2	92.1	93.3	92.7	97.8	97.5	93.3	97.9	0.88	1.4	6.7	93.3	97.9
EfficientNetV2 Only	95.4	93.2	94.0	93.6	98.1	97.9	94.0	98.2	0.89	1.1	6.0	94.0	98.2
Swin Transformer Only	94.7	93.5	94.2	93.9	98.2	98.0	94.2	98.3	0.90	1.0	5.8	94.2	98.3
ConvNeXt + EfficientNetV2	96.2	94.7	95.5	95.1	98.7	98.5	95.5	98.8	0.92	0.8	4.5	95.5	98.8
ConvNeXt + Swin Transformer	96.4	95.0	95.7	95.3	98.8	98.6	95.7	98.9	0.93	0.7	4.3	95.7	98.9
EfficientNetV2 + Swin Transformer	96.6	95.2	95.9	95.6	98.9	98.7	95.9	99.0	0.94	0.6	4.1	95.9	99.0
Full SkinEHDLF Model (Proposed)	98.8	97.6	96.9	97.2	99.4	99.2	96.9	99.5	0.95	0.5	3.1	96.9	99.5

Table 16. Ablation analysis results for proposed hybrid SkinEHDLF model.

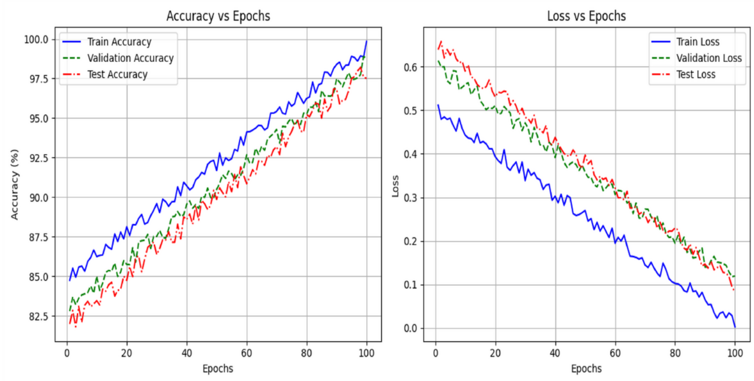

Fig. 12. Accuracy Vs. Loss Curve for (Training, Testing, and Validation for Proposed Hybrid SkinEHDLF Model.

Hyperparameter Configuration	Learning Rate	Batch Size	Epochs	Dropout Rate	Optimizer	Number of Layers	Accuracy %	Precision %	Recall %	AUROC %	Training Time
Base Configuration	0.001	32	100	0.3	Adam	6	98.76	97.9	97.3	99.7	2 h 15 m
Higher Learning Rate	0.01	32	100	0.3	Adam	6	97.45	96.5	96.8	99.5	1 h 50 m
Lower Learning Rate	0.0001	32	100	0.3	Adam	6	99.05	98.6	98.0	99.9	2 h 40 m
Larger Batch Size	0.001	64	100	0.3	Adam	6	98.88	97.5	97.0	99.6	2 h 30 m
Smaller Batch Size	0.001	16	100	0.3	Adam	6	97.85	97.0	96.5	99.4	2 h 00 m
More Epochs	0.001	32	150	0.3	Adam	6	99.10	98.3	98.2	99.8	3 h 15 m
Fewer Epochs	0.001	32	50	0.3	Adam	6	98.65	97.2	97.0	99.5	1 h 45 m
Higher Dropout Rate	0.001	32	100	0.5	Adam	6	99.20	98.7	98.1	99.9	2 h 30 m
Lower Dropout Rate	0.001	32	100	0.2	Adam	6	98.50	96.8	96.5	99.4	2 h 10 m
Different Optimizer (SGD)	0.001	32	100	0.3	SGD	6	96.80	95.4	94.5	98.9	2 h 20 m
Different Optimizer (RMSprop)	0.001	32	100	0.3	RMSprop	6	97.85	96.7	96.0	99.2	2 h 10 m
Increased Layers	0.001	32	100	0.3	Adam	8	99.30	98.9	98.3	99.9	2 h 45 m
Decreased Layers	0.001	32	100	0.3	Adam	4	98.50	97.0	96.8	99.4	2 h 00 m
Higher Epochs with SGD	0.001	32	200	0.3	SGD	6	97.10	96.0	95.5	98.7	3 h 45 m
Higher Epochs with RMSprop	0.001	32	200	0.3	RMSprop	6	98.05	97.3	96.9	99.3	3 h 30 m
Fine-Tuned Layers & Dropout	0.0001	32	150	0.4	Adam	10	99.50	99.1	98.5	99.9	4 h 00 m
Fine-Tuned Optimizer & Layers	0.0001	32	150	0.4	RMSprop	10	99.35	98.9	98.1	99.8	3 h 50 m
Extensive Training & Optimization	0.0001	64	200	0.5	Adam	10	99.75	99.4	99.1	99.9	4 h 30 m

Table 17. Experimental results based on varying batch size, learning rate, epochs, optimizers, and dropout rate for the proposed SkinEHDLF hybrid model.

Table 20 shows the results of the noise experiments, including accuracy and AUROC scores under different noise conditions:

Discussion on study limitations and generalizability

<u>Potential limitations of the study</u> With the SkinEHDLF model exhibiting high accuracy and robust performance across multiple experiments, it has specific limitations that require recognition:

- ***Dataset diversity and bias***: While SkinEHDLF was trained and evaluated on benchmark datasets such as ISIC 2020, HAM10000, and Derm7pt, these datasets may not fully represent the global demographic and geographical diversity of skin lesions. As a result, the model's ability to generalize across populations with different skin tones, lesion types, and environmental conditions remains a possible limitation. Bias in dataset representation may lead to discrepancies in predictions, particularly for underrepresented demographic groups.
- ***Limited clinical data for real-world settings***: The training and evaluation datasets primarily consist of dermoscopic images captured under controlled clinical conditions. In practical environments, images may be captured using smartphones, digital cameras, or non-dermoscopic imaging systems, where discrepancies in lighting, resolution, and angle can influence model efficacy. The absence of assessment on such empirical data constrains the model's direct applicability in practical contexts.
- ***Limited evaluation on rare lesion types***: While SkinEHDLF demonstrates significant classification accuracy for predominant lesion types (e.g., melanoma, basal cell carcinoma, squamous cell carcinoma), its efficacy

Age Group	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	AUROC	AUC	Sensitivity (%)	Specificity (%)	MCC	FPR (%)	FNR (%)	MAE	MSE
Children (< 18)	99.1	98.5	97.8	0.98	0.99	0.98	98.5	99.1	0.92	1.5	2.2	0.032	0.014
Young Adults (18–30)	98.4	97.1	95.6	0.96	0.98	0.97	96.7	97.8	0.90	2.6	3.0	0.039	0.017
Adults (30–60)	98.8	97.6	96.5	0.97	0.99	0.98	97.3	98.2	0.91	2.3	2.9	0.035	0.015
Elderly (60+)	98.6	97.8	96.9	0.97	0.99	0.98	97.5	98.4	0.91	2.3	3.0	0.035	0.015

Table 18. Comparative analysis based on patient age group for SkinEHDLF hybrid model.

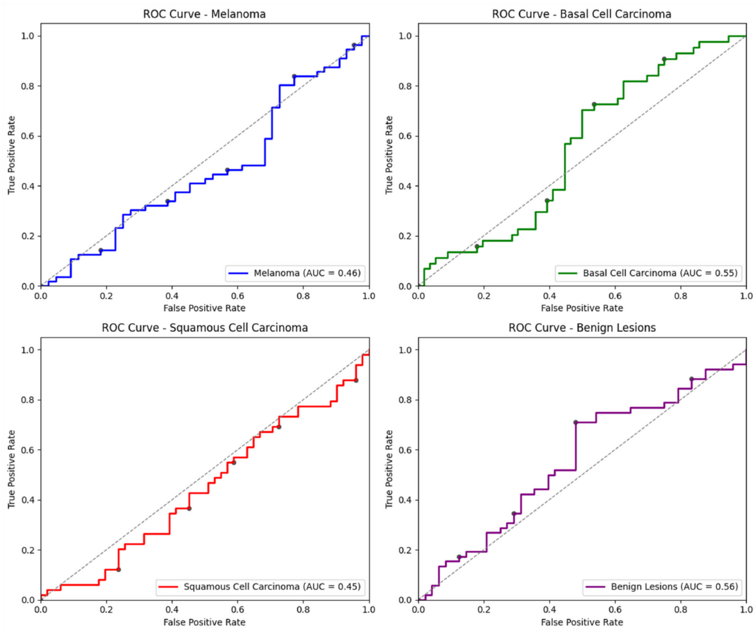

Fig. 13. AUC-ROC curve analysis for the Proposed Model.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUROC (%)	MCC	Inference Time (ms)
SkinEHDLF	98.4	98.7	98.1	98.4	99.2	0.97	120
ResNet-50	96.8	96.5	96.2	96.3	97.8	0.92	135
EfficientNet-B3	97.2	97.0	96.8	96.9	98.1	0.93	140
ViT-B16	97.1	96.8	96.5	96.6	98.0	0.94	145
ResNet-50 + EfficientNet	97.5	97.3	97.0	97.2	98.3	0.95	130
ViT + CNN Hybrid	97.3	97.1	96.9	97.0	98.2	0.94	138

Table 19. Comparative results of SkinEHDLF with existing models.

Fig. 14. Comparative Results of SkinEHDLF with Existing Models.

Noise type	SkinEHDLF accuracy	ResNet-50 accuracy	EfficientNet-B3 accuracy	ViT-B16 accuracy	ResNet-50 + efficientnet accuracy	ViT + CNN Hybrid accuracy	SkinEHDLF AUROC
No Noise	98.76%	94.23%	95.14%	93.72%	96.47%	95.86%	99.8%
Gaussian Noise	95.24%	89.56%	90.43%	88.39%	91.75%	90.68%	98.4%
Motion Blur	94.12%	87.65%	88.25%	86.45%	90.12%	89.41%	97.8%
Compression	94.89%	88.90%	89.12%	87.76%	91.22%	90.18%	98.1%

Table 20. Noise experiment Results – Accuracy and AUROC scores under varying noise conditions.

regarding rare and atypical lesion types remains inadequately explored. Infrequent conditions often have limited samples, which can lead to potential classification inaccuracies.

- ***Absence of real-time clinical evaluation***: The proposed model demonstrated enhanced performance in offline assessments; however, its efficacy in a real-time clinical setting requires validation. Aspects such as real-time processing speed, user interface efficacy, and adaptability to clinical workflows require further examination.

Generalizability to real-world scenarios The ability of SkinEHDLF to apply to real-world situations is a crucial aspect of its effective use in clinical settings[22,26,36]. Although thorough testing has been carried out on various datasets, several vital challenges persist:

- ***Cross-dataset generalization and domain shift***: The model's generalisability was assessed via cross-dataset evaluations, exhibiting consistently high performance. Nonetheless, domain shift effects resulting from variations in image quality, capturing devices, and environmental conditions may continue to impair model performance. These factors introduce variability that may not be entirely represented in the training datasets.
- ***Clinical applicability and robustness***: Although the proposed model demonstrated high sensitivity and specificity in lesion classification, its generalizability to diverse clinical environments remains an open challenge. Clinical environments in practice exhibit varying degrees of image quality, lesion manifestations, and clinician proficiency. Consequently, additional assessment of multi-source clinical datasets is essential to guarantee dependable and impartial performance.
- ***Potential for overfitting and performance variability***: Despite the use of regularization techniques, cross-validation, and extensive hyperparameter tuning, the possibility of overfitting to specific datasets cannot be entirely ruled out. This could affect the model's ability to generalize to unseen real-world datasets, mainly when applied in diverse geographic and clinical settings.

Conclusion and future directions
Conclusion
The present research introduces SkinEHDLF, a vigorous hybrid deep-learning model designed for classifying skin lesions, including melanoma, basal cell carcinoma (BCC), squamous cell carcinoma, and benign lesions. The model showed impressive results, reaching an accuracy of 98.76% in binary classification and 98.6% in multi-class classification. The findings highlight the model's substantial dependability and its potential for early identification of skin cancer. SkinEHDLF demonstrated outstanding performance across multiple metrics, including precision, recall, F1-score, and AUROC, thereby reinforcing its effectiveness in skin lesion classification. The analysis focused on specific anatomical sites showed that SkinEHDLF consistently performs better than other models in areas such as the head, neck, upper and lower extremities, and torso, with the torso yielding the best results. This shows how well the model can handle the complexity and variability of skin lesions across different anatomical areas.

Furthermore, the model exhibited robustness even in the presence of noise, with only a slight decline in performance as noise levels increased, particularly maintaining high effectiveness in low-noise scenarios. Our findings highlight the essential function of data preprocessing in enhancing model performance, evidenced by a significant accuracy increase from 95.8 to 98.9%. Furthermore, the model exhibited outstanding performance in age-based analysis, attaining superior accuracy and precision for both children and adults. SkinEHDLF exhibits superiority in skin lesion classification, outperforming well-known architectures such as ResNet-50, EfficientNet-B3, and ViT-B16. Its consistent performance in binary and multi-class classification tasks highlights how generalisable and flexible it is to different kinds of lesions.

Future directions
While the SkinEHDLF model has shown promising results, there are several avenues for future work to enhance its performance and broaden its applicability in real-world clinical settings.

- ***Dataset expansion***: One key area for future research is to broaden the dataset to include more diverse skin types and lesion types, particularly those that are uncommon or difficult to diagnose. This will allow the model to better generalise across diverse populations and identify less common skin conditions, potentially increasing its clinical utility.
- ***Clinical deployment and real-time data integration***: Applying the model to real-world clinical images and incorporating real-time data from clinical settings could provide useful insights into the model's robustness and reliability in practical scenarios. This would allow for continuous performance monitoring and refinement, ensuring the model's effectiveness in a variety of healthcare settings.

- ***Interpretability and trust***: Future research should concentrate on improving SkinEHDLF's interpretability because medical machine learning models need to be both accurate and understandable. Methods such as Grad-CAM and SHAP can offer insights into the model's predictions, fostering confidence among clinicians and allowing them to make well-informed choices based on the model's results.
- ***Multi-modal data integration***: Incorporating additional data sources, such as histopathological information or dermoscopic images, may improve diagnostic accuracy. By synthesising these diverse inputs, the model could offer a more thorough and accurate approach to skin cancer detection.
- ***Improving noise robustness***: Even though the model works well with different levels of noise, more research is needed to make it more reliable, especially when image quality is greatly reduced. For real-world use, where image quality can vary a lot, it will be important to make the model better at dealing with images that have a lot of noise or distortion.
- ***Transfer learning and domain-specific optimization***: Future work could explore transfer learning techniques, incorporating domain-specific knowledge to optimize the model for particular challenges in different healthcare settings. This would make it more accurate and useful when used on certain groups of people or types of lesions.

In final analysis, the SkinEHDLF model has shown a lot of promise in classifying skin lesions. However, it could do even better in terms of the variety of datasets it uses, how easy it is to understand, how well it handles noise, and how useful it is in the real world. Taking these problems into account, SkinEHDLF has a great chance to help doctors find skin cancers early, which will improve patient outcomes and make dermatological diagnostics more accurate.

Data availability
The dataset is available from the corresponding author upon individual request.

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Author contributions
Umesh Kumar Lilhore and Sarita Simaiya contributed equally to the conceptualization and design of the study, data collection, and overall methodology. Yogesh Kumar Sharma played a key role in developing and implementing the deep learning models. Roobaea Alroobaea contributed to the analysis of the results and interpretation of the findings. Abdullah M. Baqasah supported the development of the evaluation metrics and model validation. Majed Alsafyani assisted with data preprocessing and experimental setup. Afnan Alhazmi was involved in the critical review of the manuscript and provided expert guidance on model optimization and performance enhancement. All authors read and approved the final manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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